Two-view geometry



Epipolar geometry

3D reconstruction

F-matrix comp.

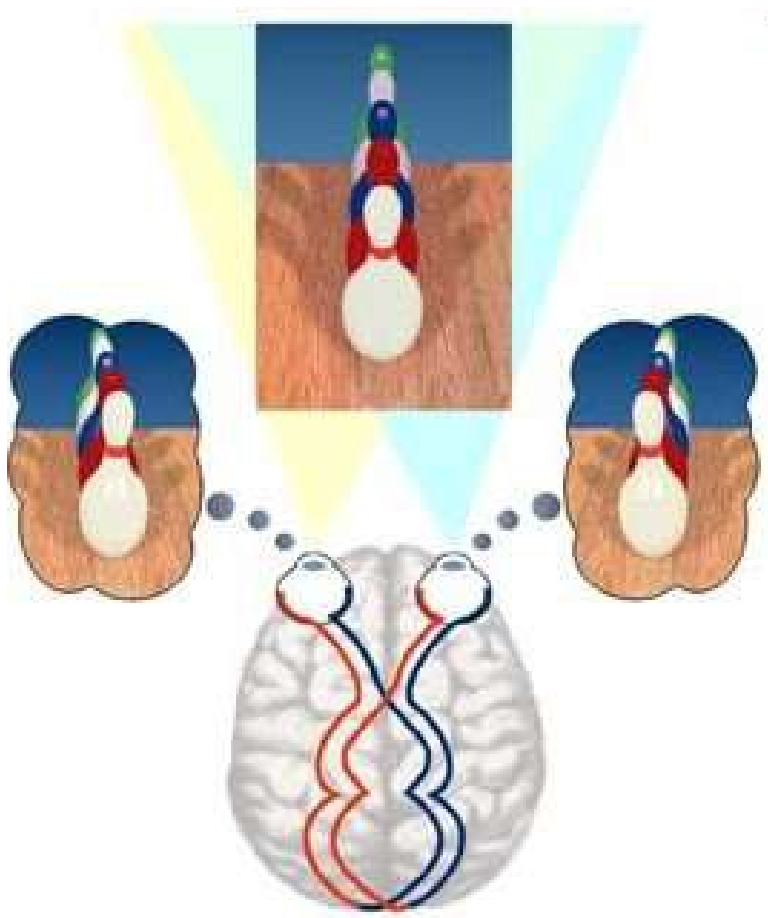
Structure comp.

Intrisic ambiguity in 3D world → 2D image projection



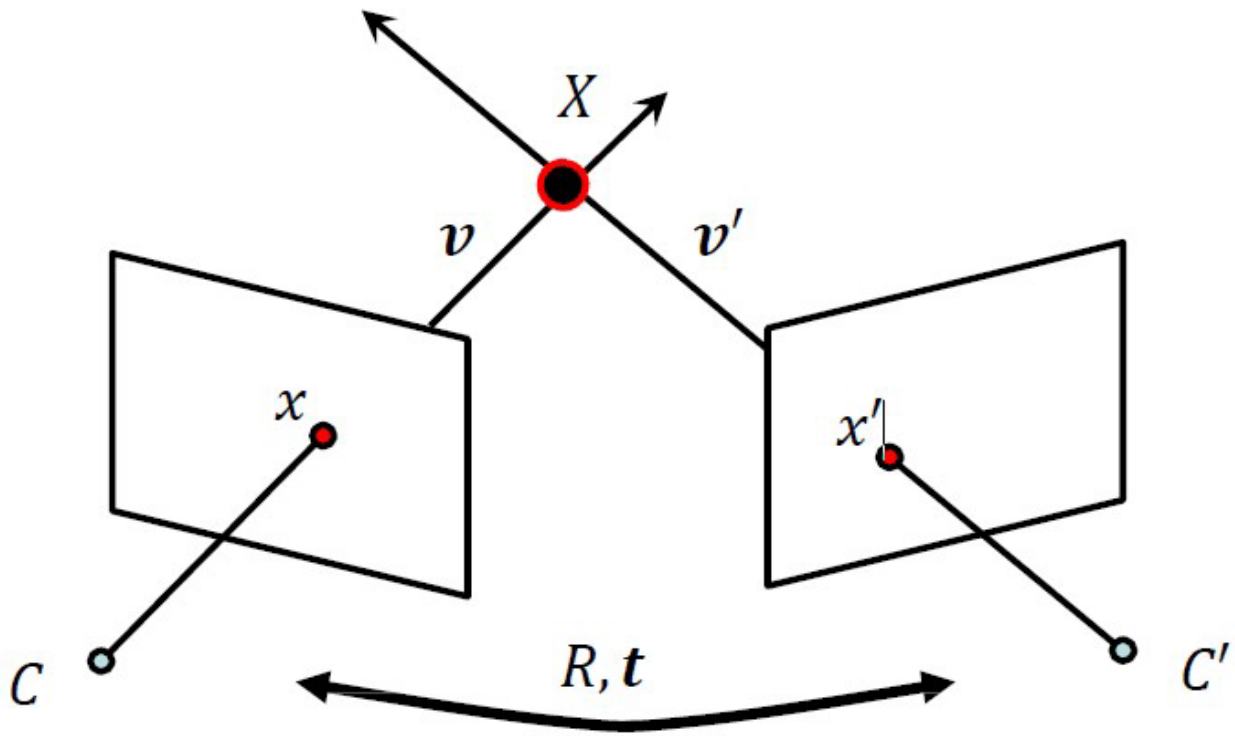
Courtesy slide S.Lazebnik

After projection, different depths cannot be distinguished in the image.

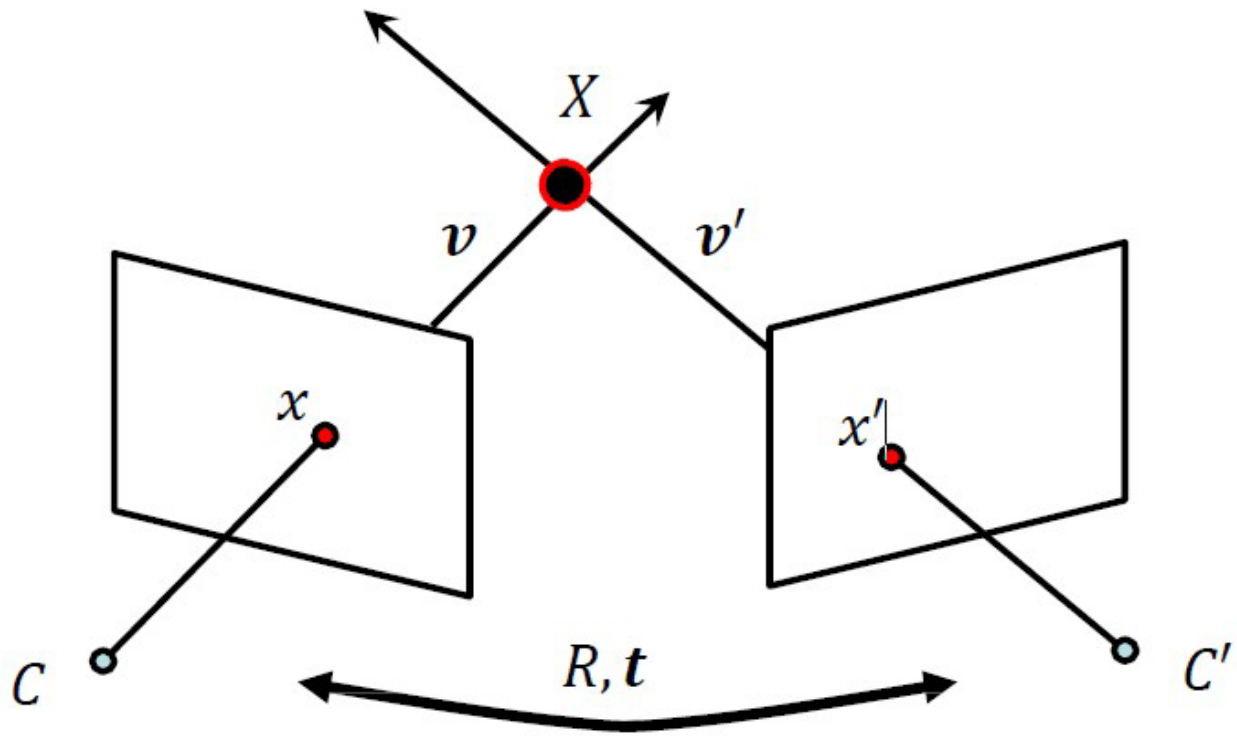


Multiple views of the same scene can help to solve such ambiguities:
In images taken from different viewpoints, ambiguities are not the same

Multiple views of a point X in 3D



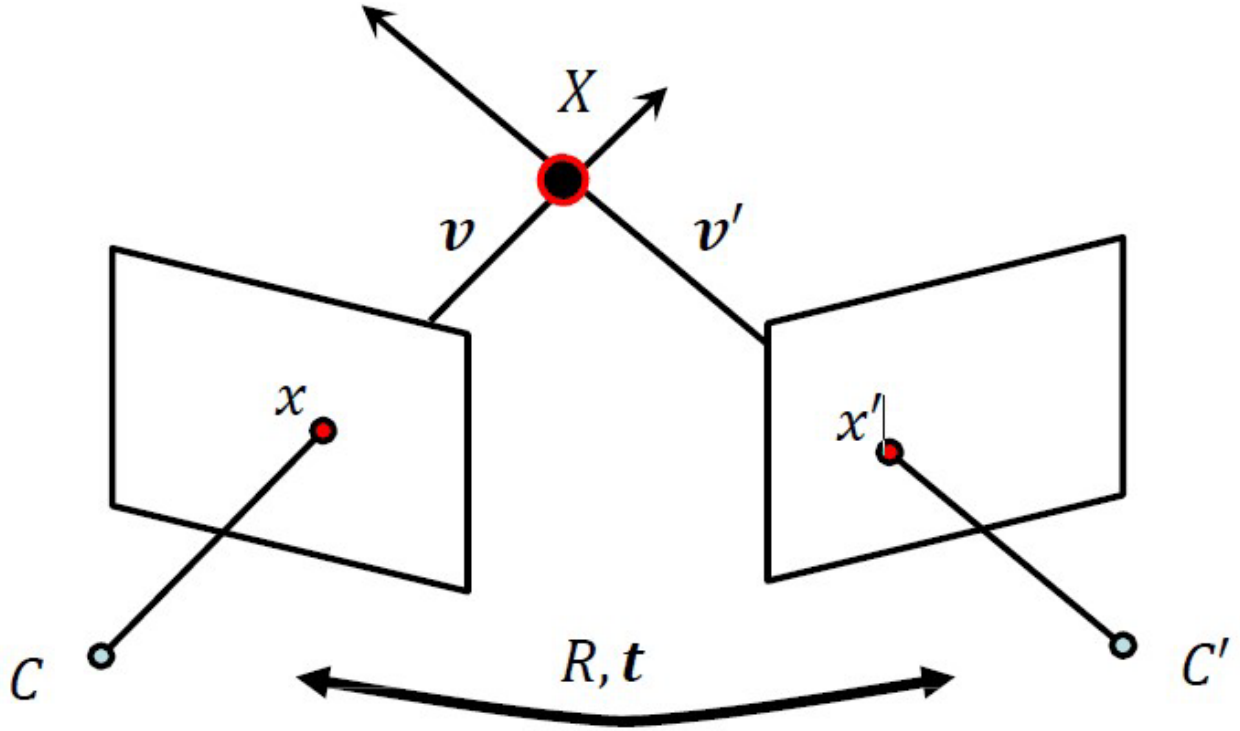
Multiple views of a point X in 3D: scenarios



scenario 1: STEREO vision

x ←→x’ known; R,t known; K and K’ known;

→ find **X**



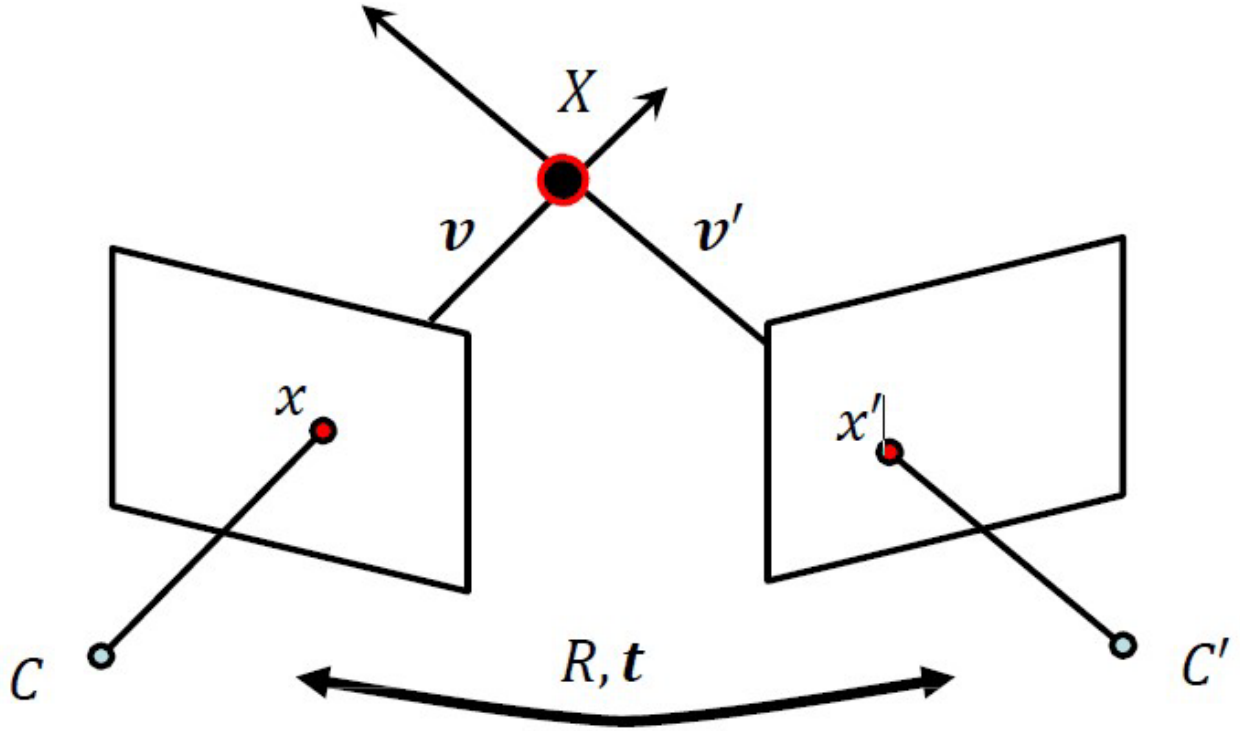
from K(K’) and x(x’) we compute viewing ray $\boldsymbol{v}(\boldsymbol{v}')$ →

triangulation: $\boldsymbol{X} = \boldsymbol{v} \cap \boldsymbol{v}'$

scenario 2: calibrated structure from motion

x ←→x’ known; **R,t unknown**; K and K’ known;

→ find **X** and **R,t**



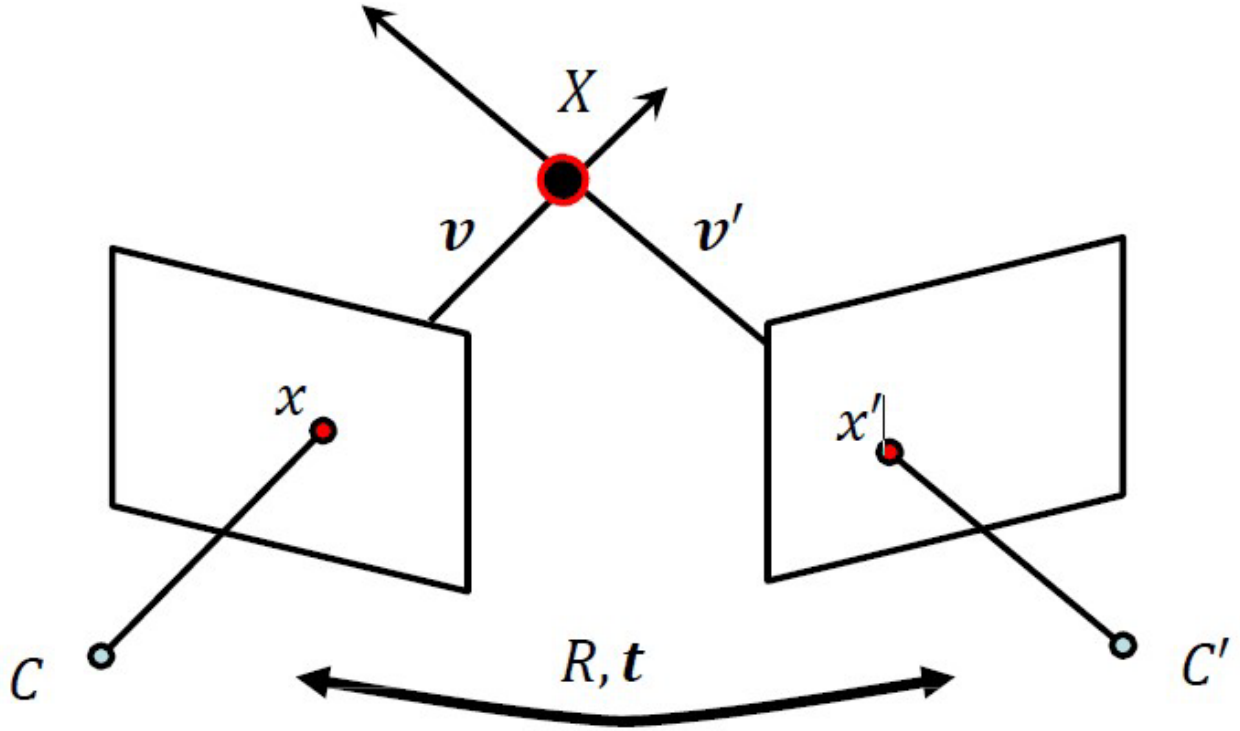
use **epipolar constraint** to estimate R, t; → compute viewing rays $\boldsymbol{v}$, $\boldsymbol{v}'$

→ triangulation: $\boldsymbol{X} = \boldsymbol{v} \cap \boldsymbol{v}'$

scenario 3: uncalibrated structure from motion

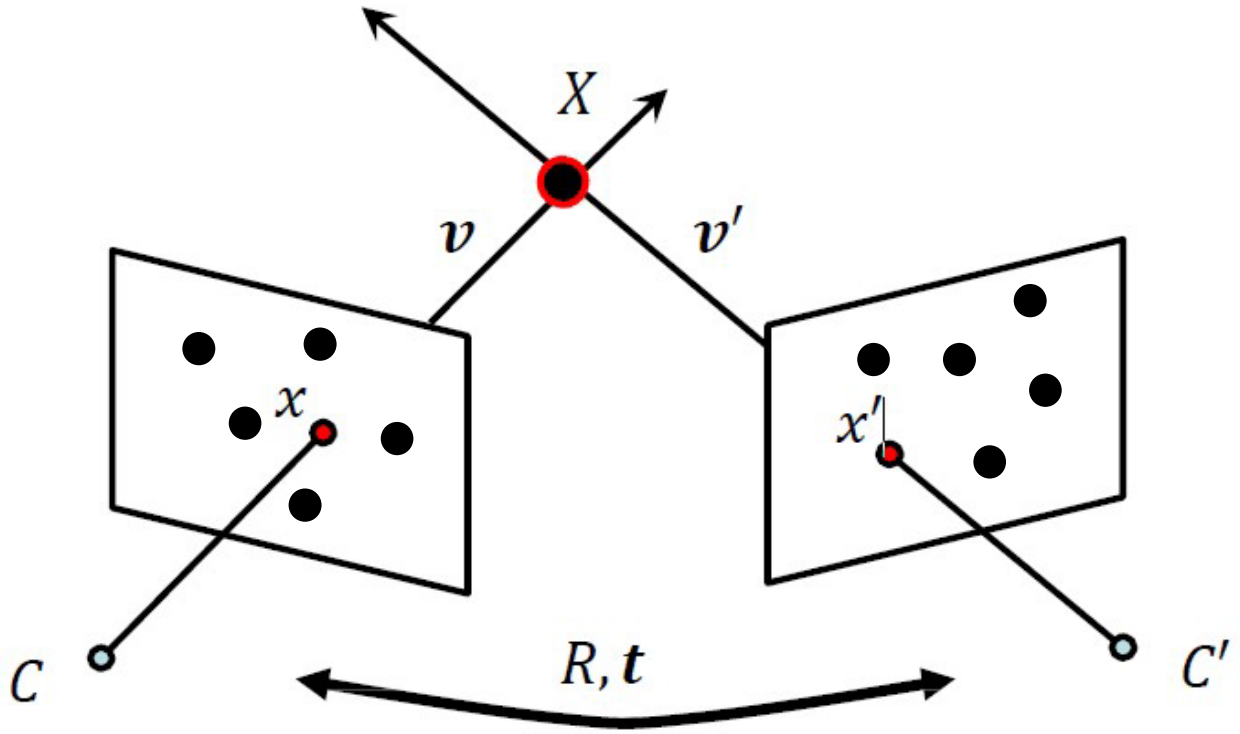
x ←→x’ known; **R,t unknown**; **K and K’ unknown**;

→ find X, **R,t** and **K, K’**



use **epipolar constraint** & partial information on scene and/or cameras to estimate K, K’, R, t; then compute viewing rays $\boldsymbol{v}$, $\boldsymbol{v}'$ → triangulation: $\boldsymbol{X} = \boldsymbol{v} \cap \boldsymbol{v}'$

First of all: given many features in both images find pairs of **corresponding** features



Feature extraction, feature descriptor computation and feature matching

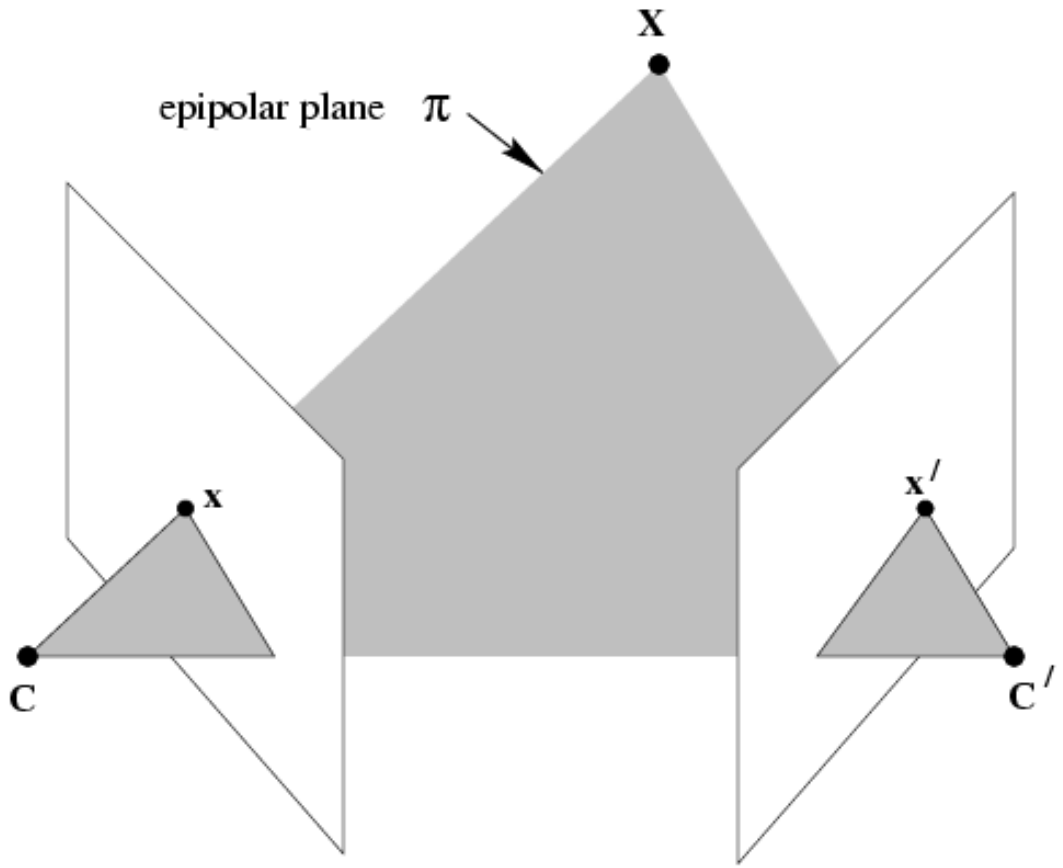
Three questions:

(i) **Correspondence geometry:** Given an image point $\mathbf{x}$ in the first view, how does this constrain the position of the corresponding point $\mathbf{x}'$ in the second image?

(ii) **Camera geometry (motion):** Given a set of pairs of corresponding image points $\{x_i \leftrightarrow x'_i\}$, i=1,…,n, what are the cameras P and P' for the two views?

(iii) **Scene geometry (structure):** Given corresponding image points $x_i \leftrightarrow x'_i$ and cameras P, P', what is the position of (their backprojection) X in space?

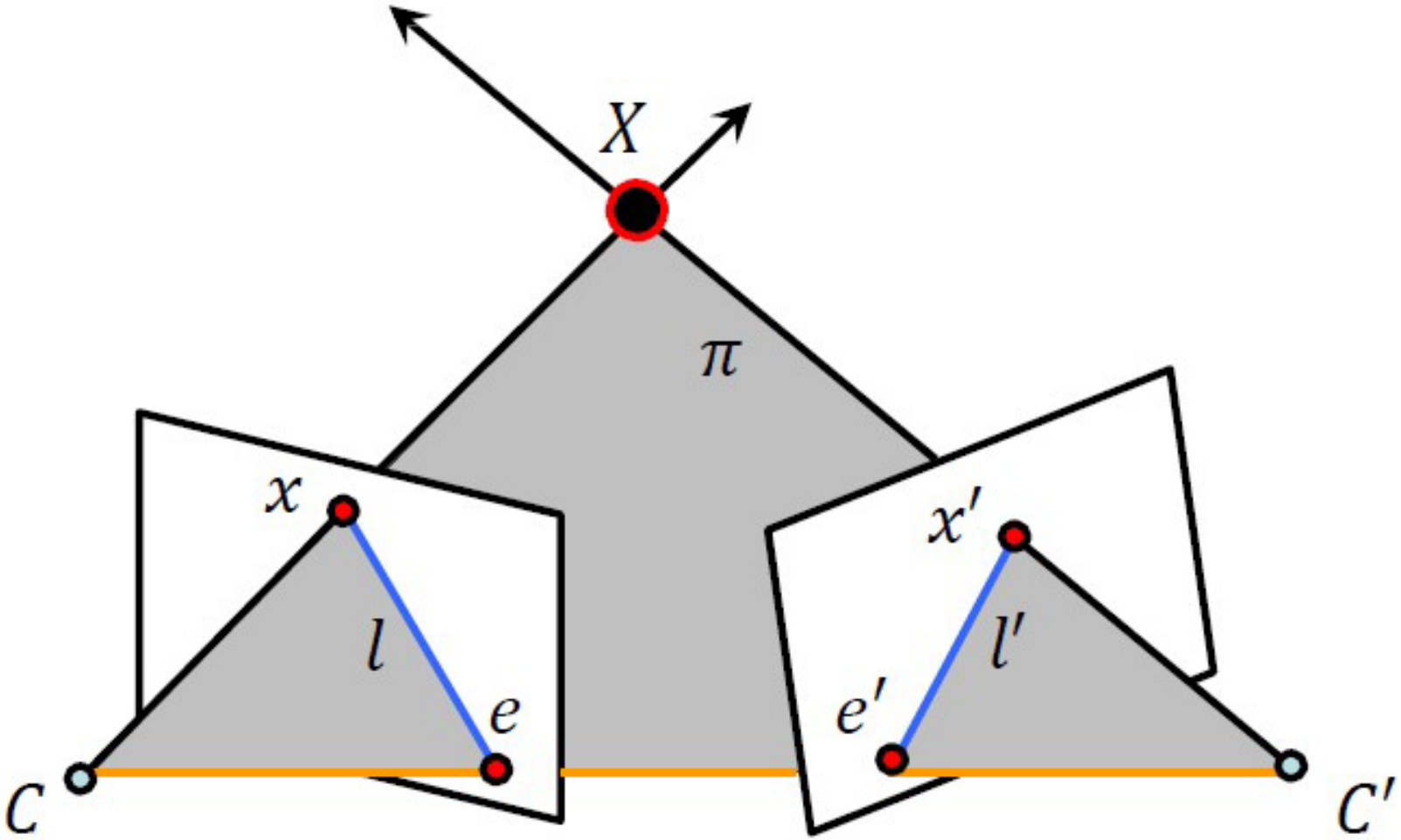
The epipolar geometry



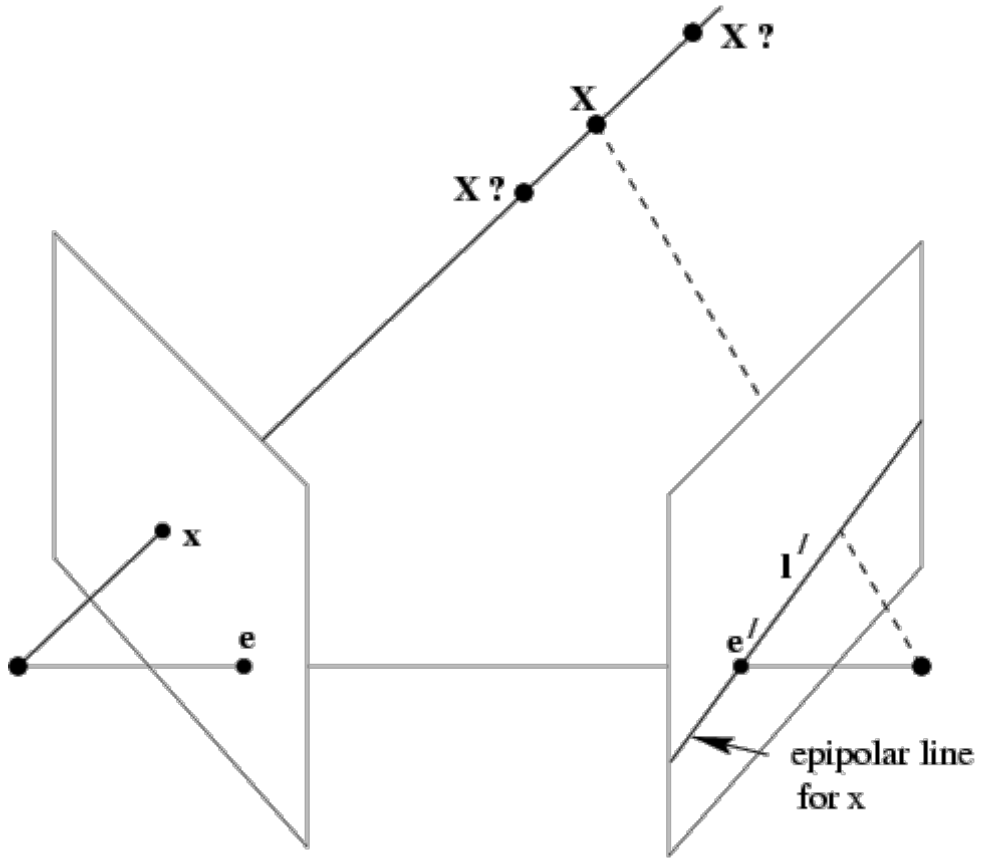
Scene point, its images and camera centers are coplanar

C,C',x,x' and X are coplanar

The epipolar constraint: viewing rays of corresponding image points must intersect in 3D space

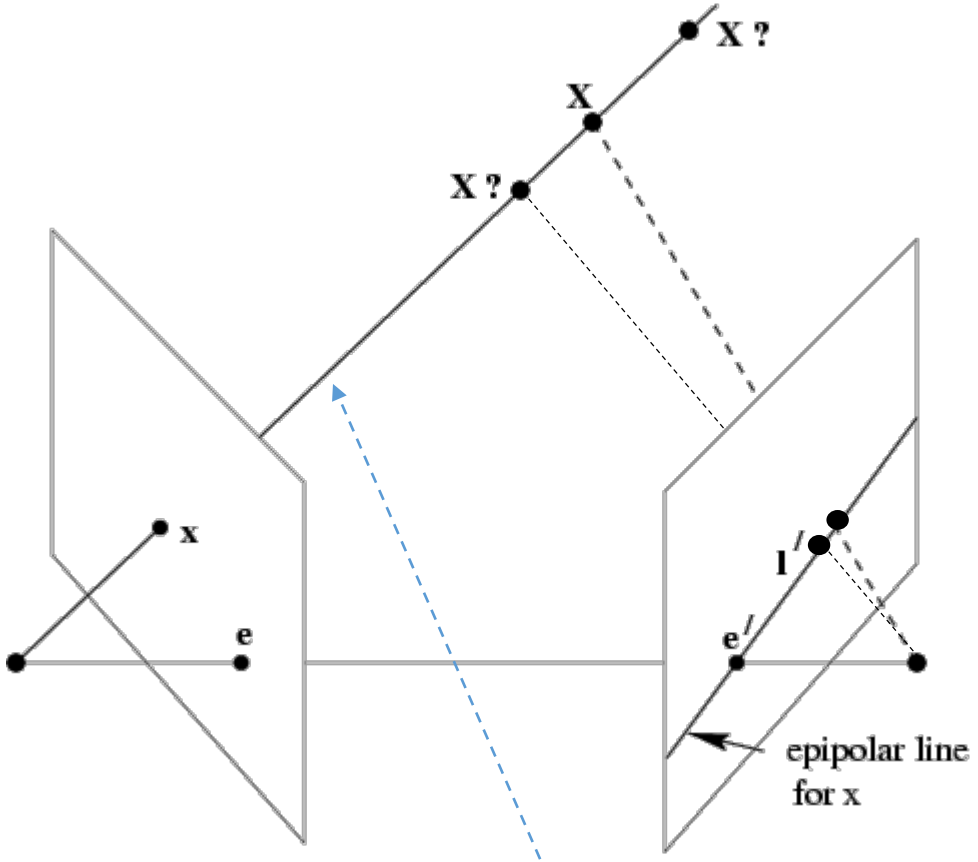


The epipolar constraint: viewing rays of corresponding points must intersect in 3D space



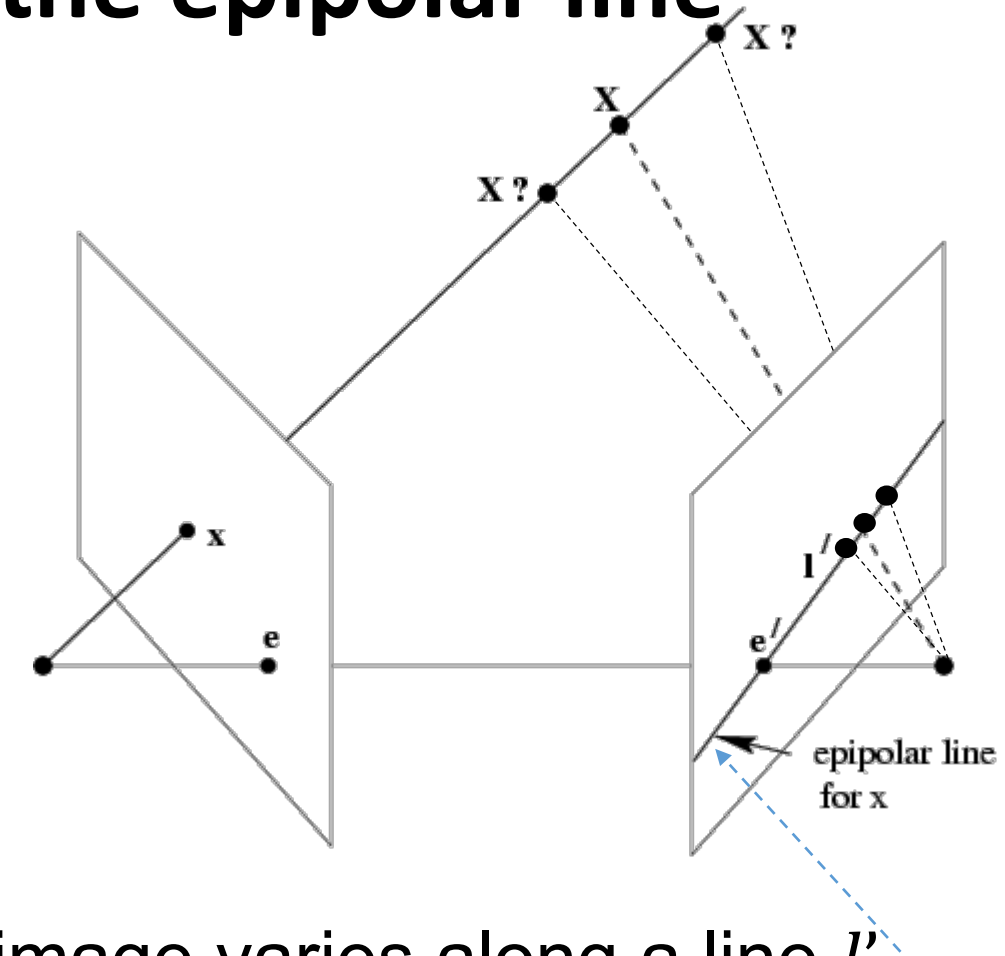
What if only C,C’,x are known?

The epipolar constraint



possible position of X in the 3D space
varies along the viewing ray associated to x

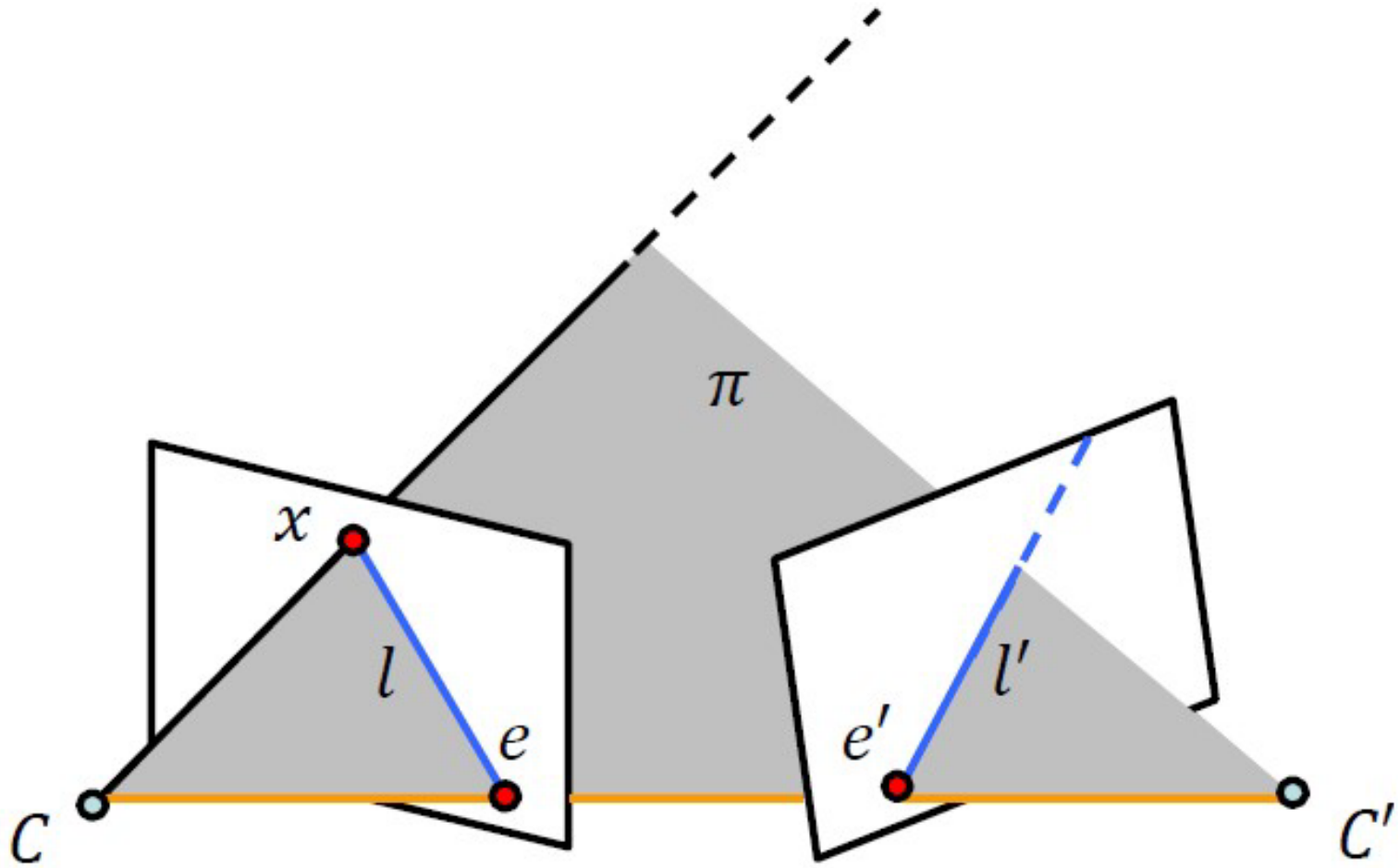
The epipolar constraint: the epipolar line



their image varies along a line l'.

l': image projection of the viewing ray

Correspondence between epipolar lines



The epipolar constraint: the epipole

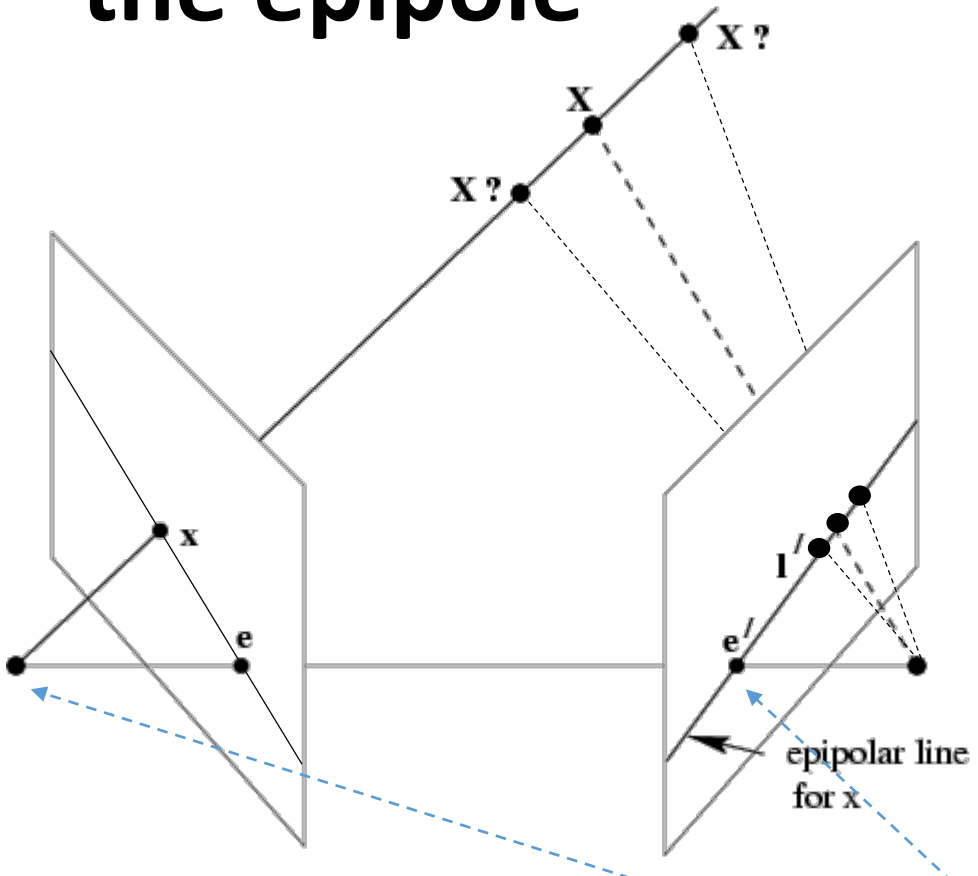
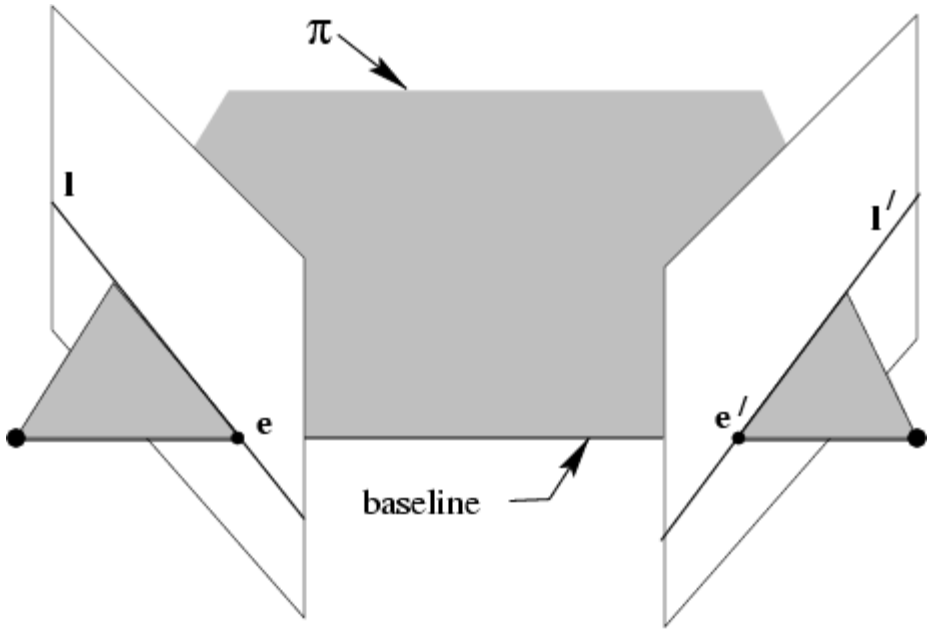


image projection of the viewing ray

the viewing ray is a line through the 1st camera center

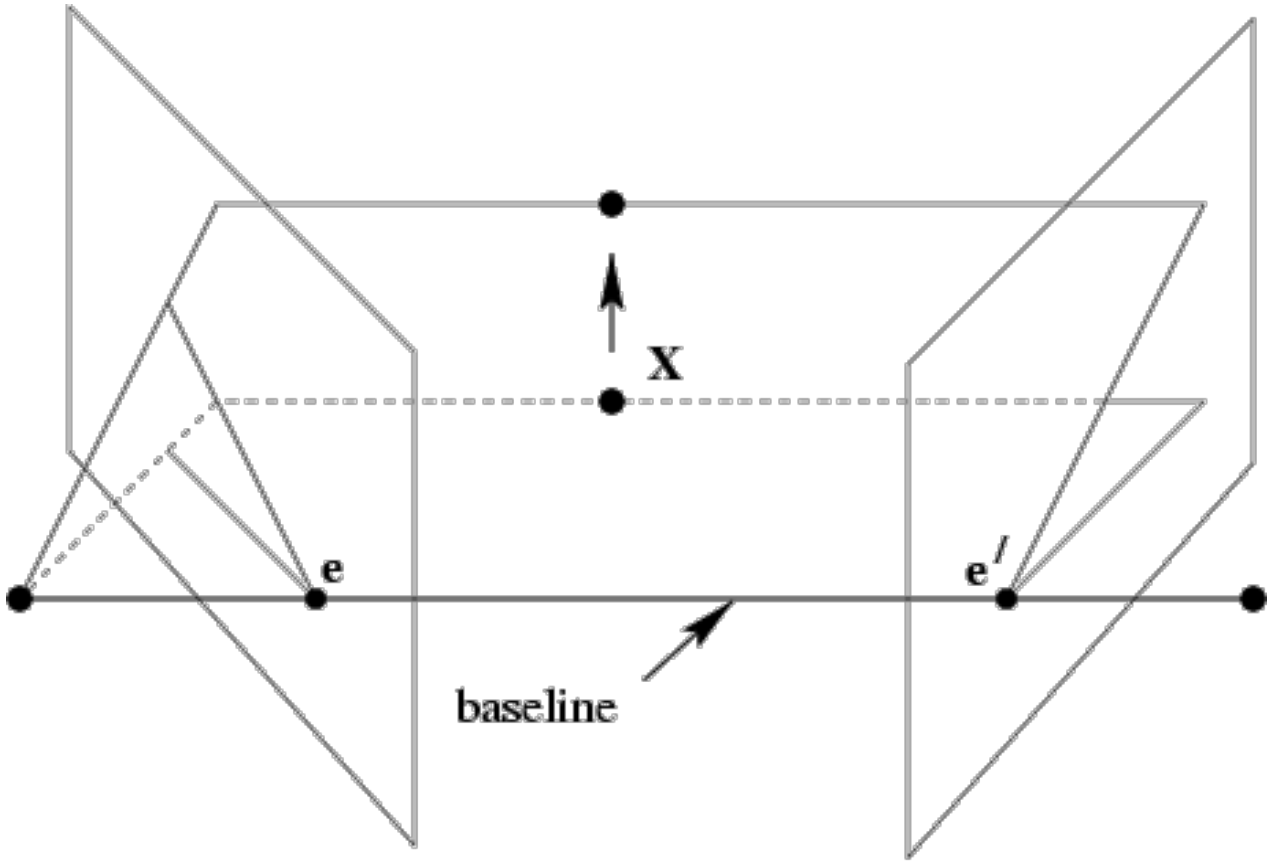
→its image l' always goes through the image $\mathbf{e}'$ of the 1st camera center

The epipolar constraint



All points on π project on l and l'

The epipolar constraint

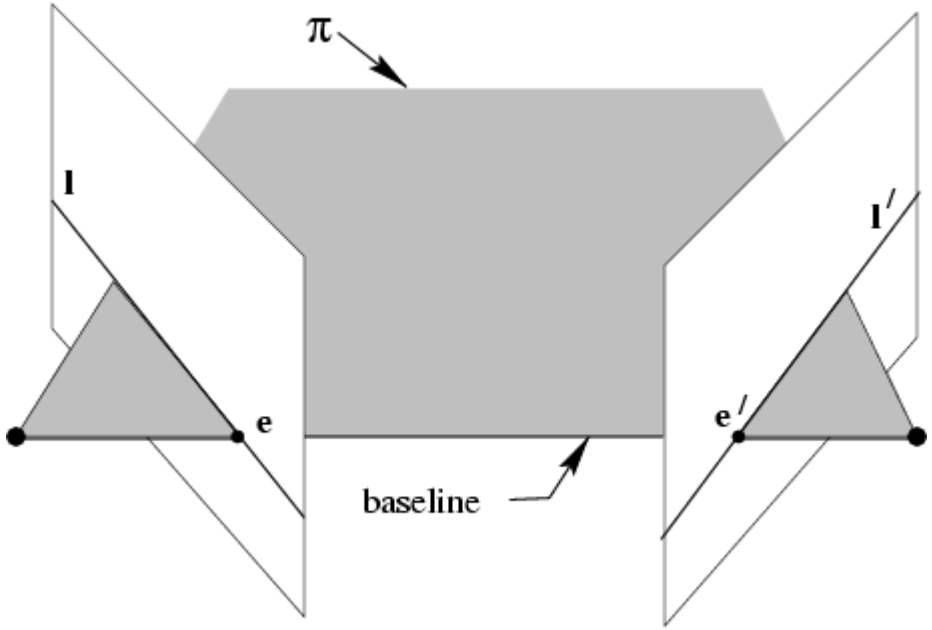


Family of coaxial planes π (with baseline as axis) cross the image planes at families of epipolar lines l and l'

Family of lines l concur at e and family of lines l' concur at e'

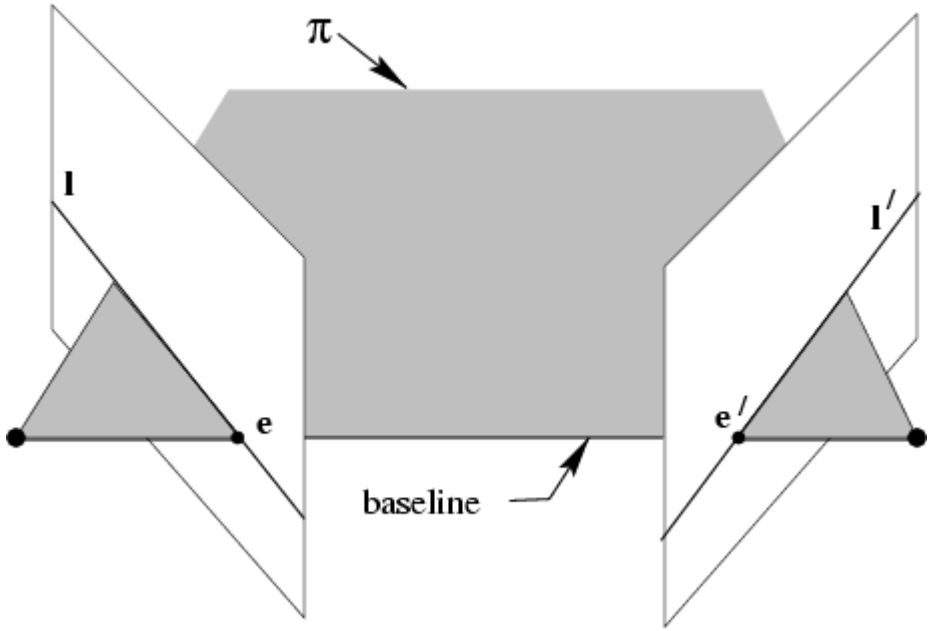
Epipoles and epipolar lines

epipoles e,e’
= intersection of baseline with image plane
= projection of projection center in other image
= vanishing point of camera motion direction



an epipolar plane = plane containing baseline (1-D family)

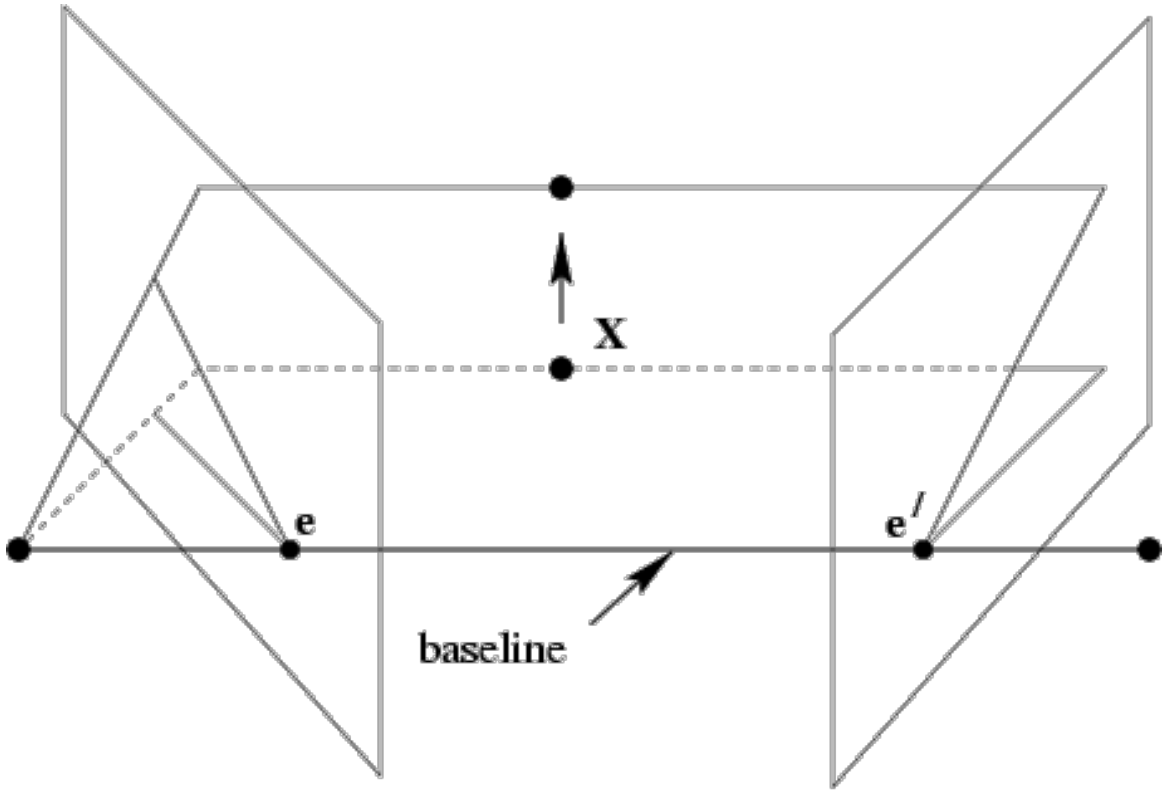
an epipolar line = intersection of epipolar plane with image
(always come in corresponding pairs)



an epipolar plane = plane containing baseline (1-D family)

an **epipolar line** = intersection of epipolar plane with image plane (always come in corresponding pairs), goes through epipole

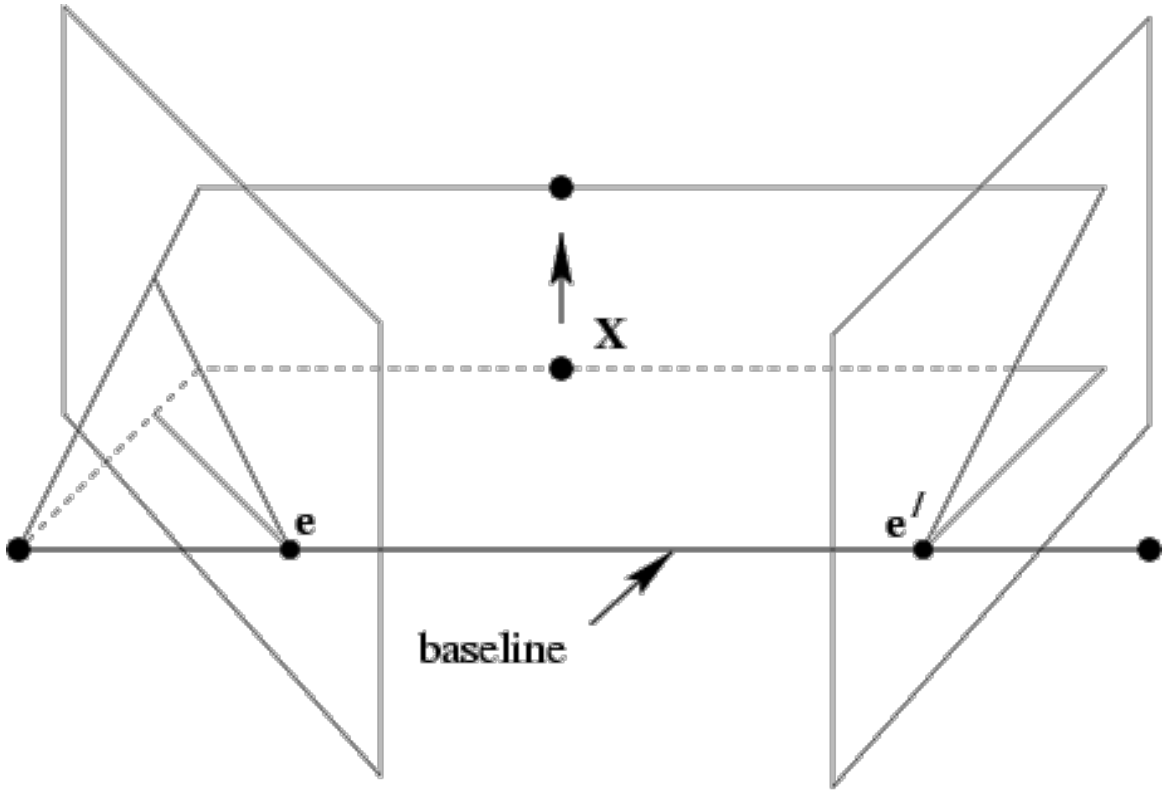
→ 1-D family of coplanar, concurrent lines
→ projectively related (same cross ratio in the two images)



→ **epipolar lines**: 1-D family of coplanar, concurrent lines

→ projectively related (same cross ratio in the two images = 3D cross ratio of the epipolar planes)

→ concur at the **epipoles**



epipole e,e’

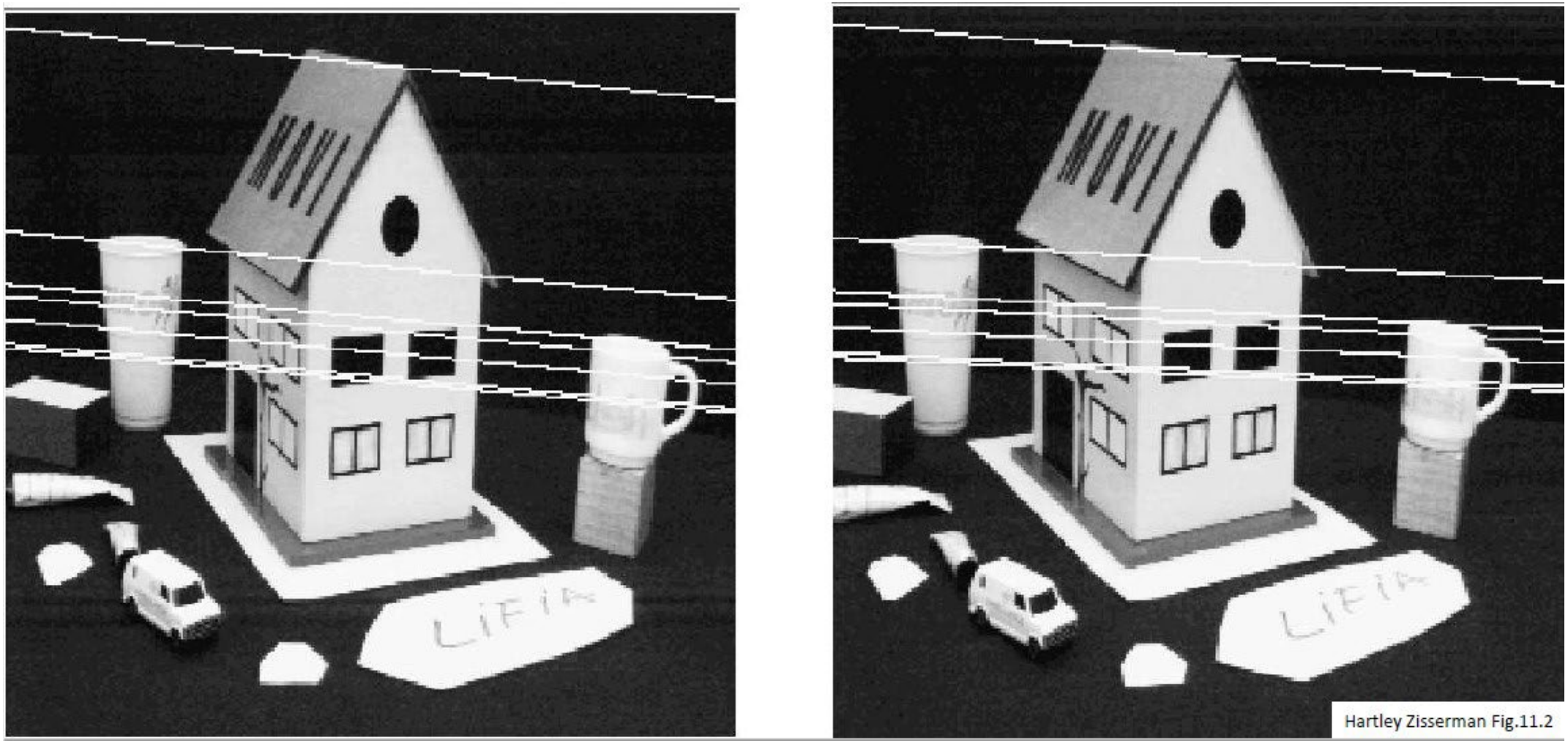
= intersection of baseline with image plane

= projection of camera center in other image

= vanishing point of camera relative motion direction

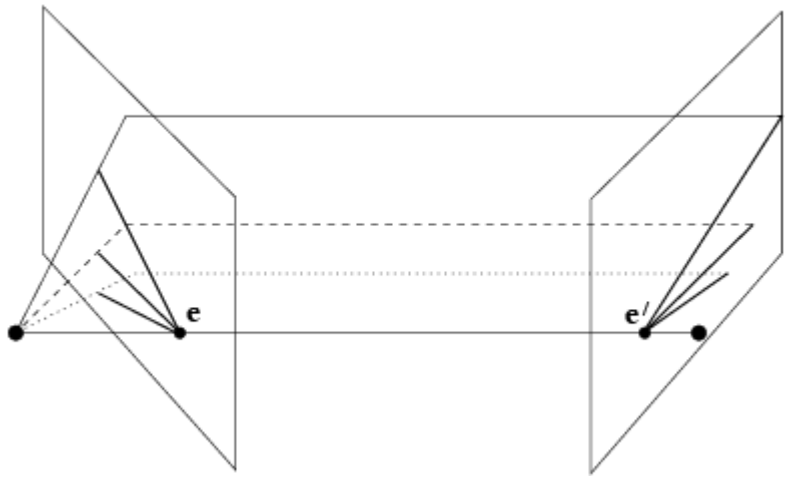
= concurrency point for epipolar lines

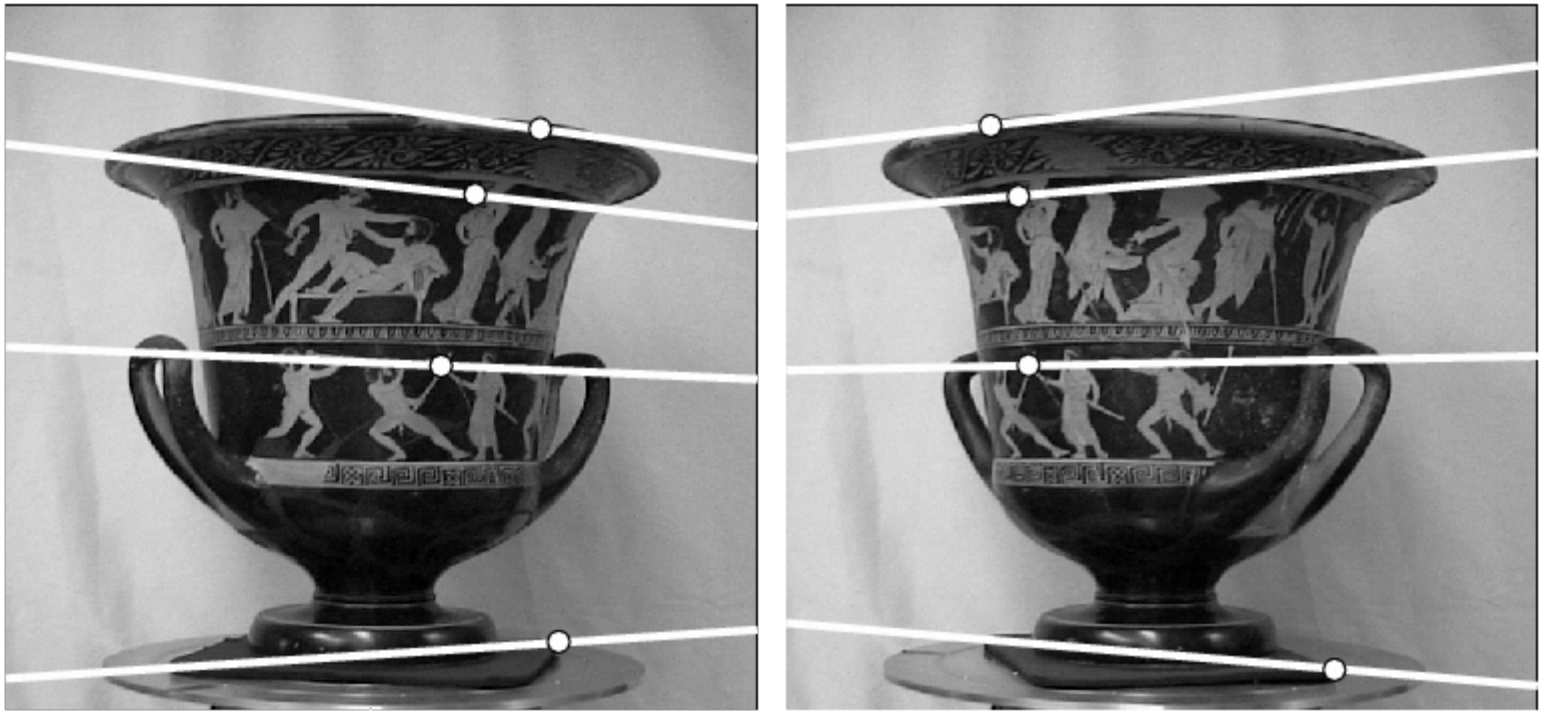
Epipolar lines



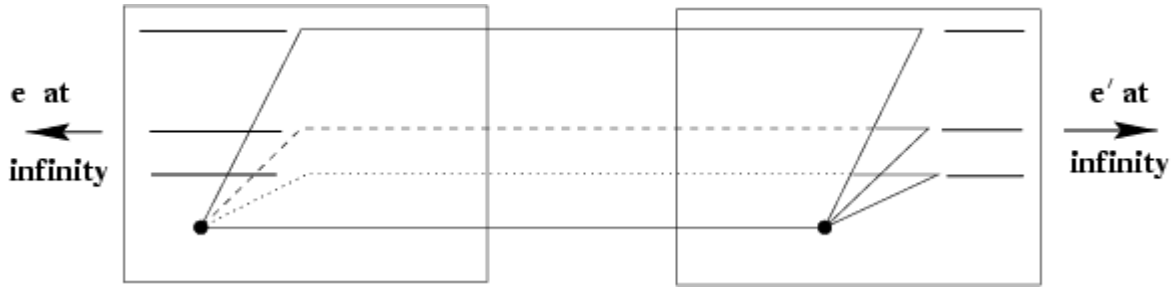
Hartley Zisserman Fig.11.2

Example: converging cameras





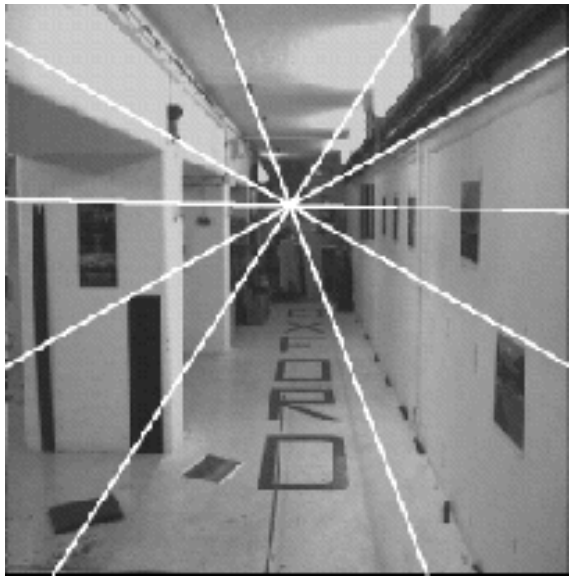
Example: motion parallel to image plane

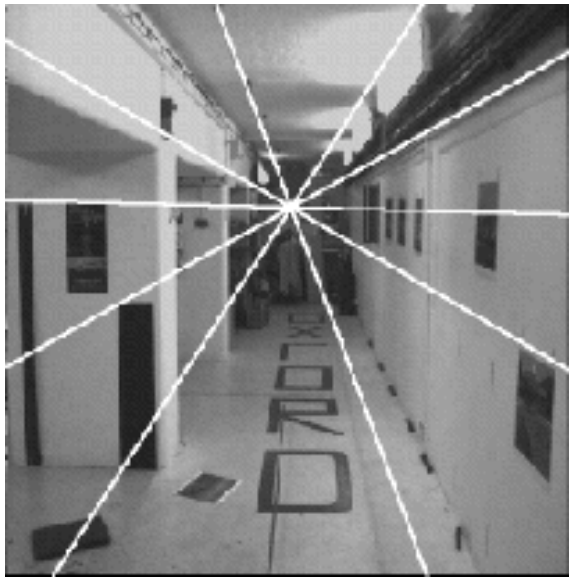


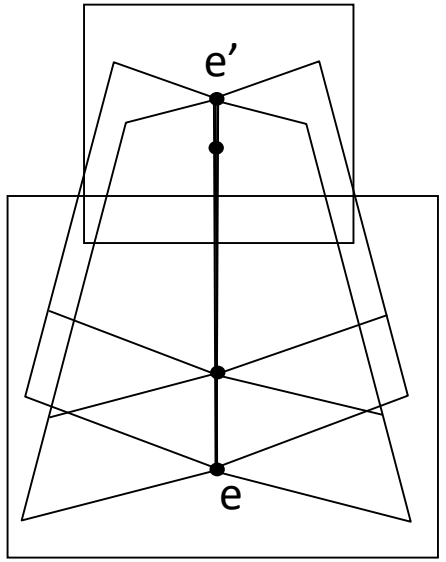




Example: forward motion (translation)





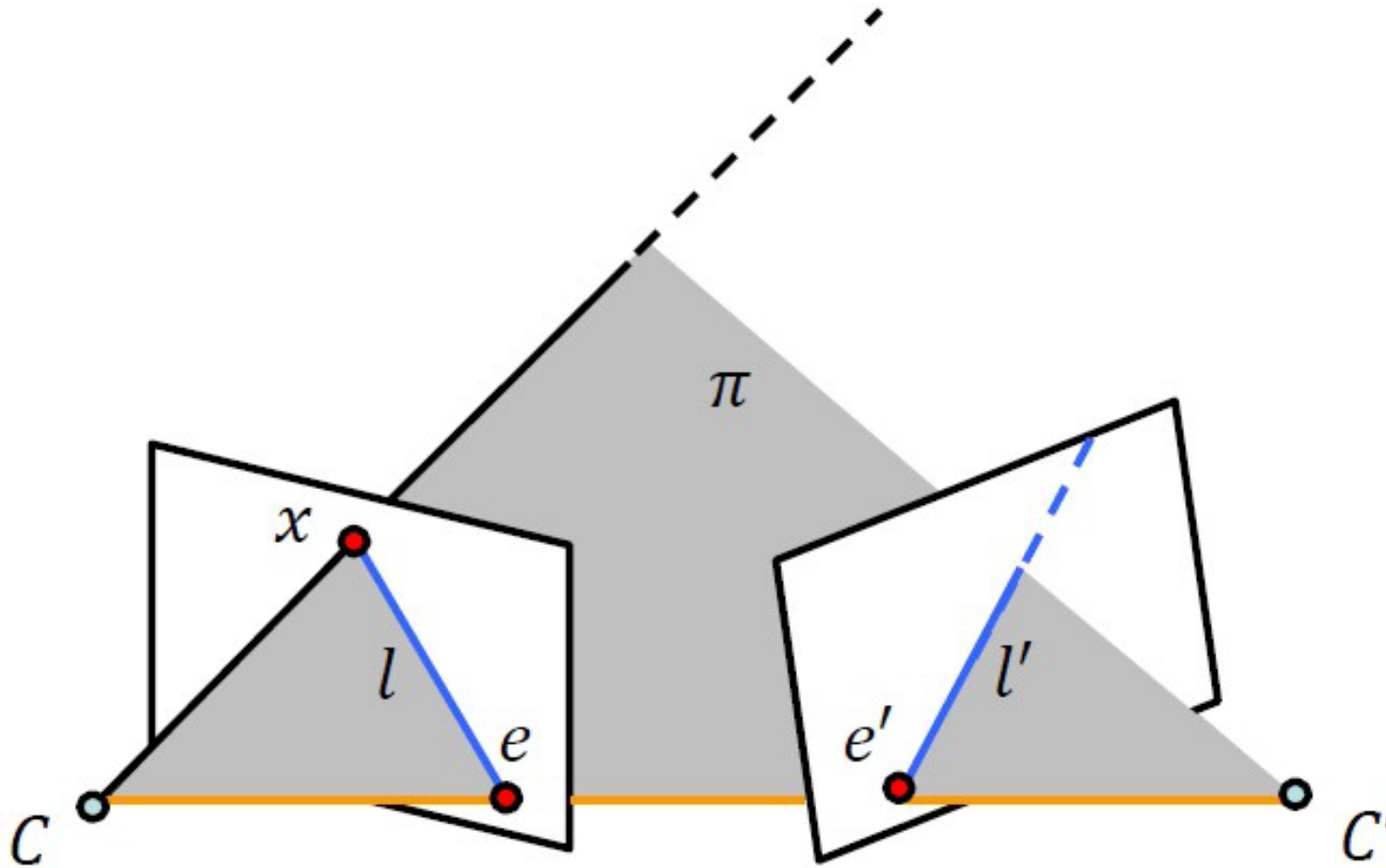


The fundamental matrix F (Faugeras, 1992)

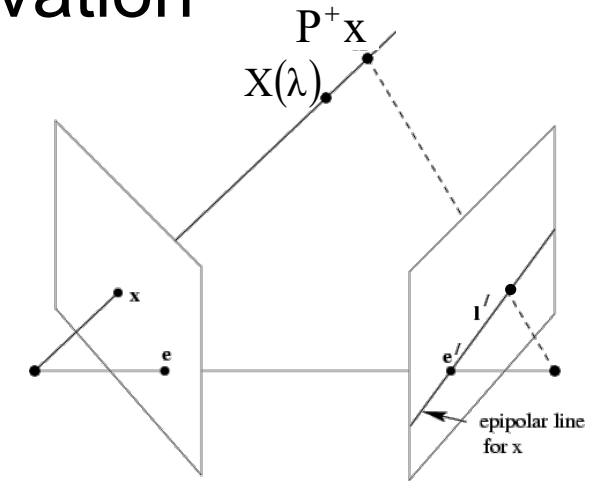
algebraic representation of epipolar geometry

$$x \mapsto l' = Fx$$

we will see that mapping is (singular) correlation
(i.e. projective mapping from points to lines)
represented by the fundamental matrix F


π
x
l
e
C
e′
l′
C′

The fundamental matrix F derivation



First image of a point X

$$x = PX = [M|m]\,X$$

Locus of points X: viewing ray through x

$$X = O + \lambda \begin{bmatrix} M^{-1}x \\ 0 \end{bmatrix}$$

Project locus of X onto second camera P′

$$x' = P'X = P'O + \lambda [M'|m'] \begin{bmatrix} M^{-1}x \\ 0 \end{bmatrix} = e' + \lambda M'M^{-1}x$$

l' is the image of the viewing ray through x

$$x' \in l' = e' \times M'M^{-1}x = [e']_{\times} M'M^{-1}x$$

skew-symmetric matrix used to compute a cross-product by means of matrix multiplication

$$\boldsymbol{a} \times \boldsymbol{b} = \begin{bmatrix} 0 & -a_z & a_y \\ a_z & 0 & -a_x \\ -a_y & a_x & 0 \end{bmatrix} \begin{bmatrix} b_x \\ b_y \\ b_z \end{bmatrix}$$

Thus the matrix associated to the cross product against $\boldsymbol{e}'$ is

$$[\boldsymbol{e}']_\times = \begin{bmatrix} 0 & -e'_z & e'_y \\ e'_z & 0 & -e'_x \\ -e'_y & e'_x & 0 \end{bmatrix}$$

$$\mathrm{x}' \in l' = F\,\mathrm{x} = [\mathrm{e}']_{\times}\mathrm{M}'\mathrm{M}^{-1}\mathrm{x}$$

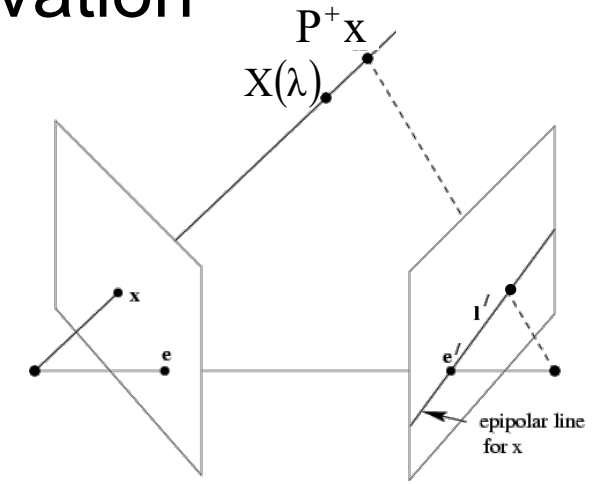
where

$$[\mathrm{e}']_{\times} = \begin{bmatrix} 0 & -\mathrm{e}'_z & \mathrm{e}'_y \\ \mathrm{e}'_z & 0 & -\mathrm{e}'_x \\ -\mathrm{e}'_y & \mathrm{e}'_x & 0 \end{bmatrix}$$

is a skew-symmetric matrix used to compute a cross-product by means of matrix multiplication

$[\mathrm{e}']_{\times}$ is singular

The fundamental matrix F derivation



First image of a point X

$$\mathrm{x} = \mathrm{PX} = [\mathrm{M}|\mathrm{m}]\,\mathrm{X}$$

Locus of points X: viewing ray through x

$$\mathrm{X} = \mathrm{O} + \lambda \begin{bmatrix} \mathrm{M}^{-1}\mathrm{x} \\ 0 \end{bmatrix}$$

Project locus of X onto second camera P′

$$\mathrm{x}' = \mathrm{P}'\mathrm{X} = \mathrm{P}'\mathrm{O} + \lambda[\mathrm{M}'|\mathrm{m}'] \begin{bmatrix} \mathrm{M}^{-1}\mathrm{x} \\ 0 \end{bmatrix} = \mathrm{e}' + \lambda \mathrm{M}'\mathrm{M}^{-1}\mathrm{x}$$

l' is the image of the viewing ray through x

$$\mathrm{x}' \in l' = \mathrm{e}' \times \mathrm{M}'\mathrm{M}^{-1}\mathrm{x} = [\mathrm{e}']_{\times}\mathrm{M}'\mathrm{M}^{-1}\mathrm{x}$$

x′ is the 2nd image of X

$$\mathrm{x}' \in l' \Rightarrow \mathrm{x}'^{T} l' = 0 \rightarrow \mathrm{x}'^{T}[\mathrm{e}']_{\times}\mathrm{M}'\mathrm{M}^{-1}\mathrm{x} = 0$$

$$\mathrm{x}'^{T} F\, \mathrm{x} = 0 \qquad \text{with} \qquad F = [\mathrm{e}']_{\times}\mathrm{M}'\mathrm{M}^{-1}$$

The fundamental matrix $F = [e']_{\times} M' M^{-1}$

correspondence condition

The fundamental matrix satisfies the condition that for any pair of corresponding points $x \leftrightarrow x'$ in the two images $x'^{T} F x = 0 \quad \left(x'^{T} l' = 0\right)$

even if cameras are not calibrated, i.e., P and P' are unknown, the fundamental matrix F can still be computed by imposing correspondence conditions: $x'^{T} F x = 0$

The fundamental matrix F: properties

- $F = [e']_{\times} M' M^{-1}$ is the only 3x3 rank 2 matrix satisfying $x'^{T} F x = 0$ for all $x \leftrightarrow x'$

- **Transpose:** if F is fundamental matrix for (P,P'), then F^{T} is fundamental matrix for (P',P)

- **Epipolar lines:** $l' = F x$ & $l = F^{T} x'$

- **Epipoles are LNS of F:** on all epipolar lines, thus $e'^{T} F x = 0$, $\forall x \Rightarrow e'^{T} F = 0$, similarly $Fe = 0$

- **F** has 7 d.o.f. , i.e. 3x3 -1(homogeneous) -1(rank-2 constraint)

- **F** is a correlation, projective mapping from a point x to a line l'=Fx (not a proper correlation, because it is not invertible)

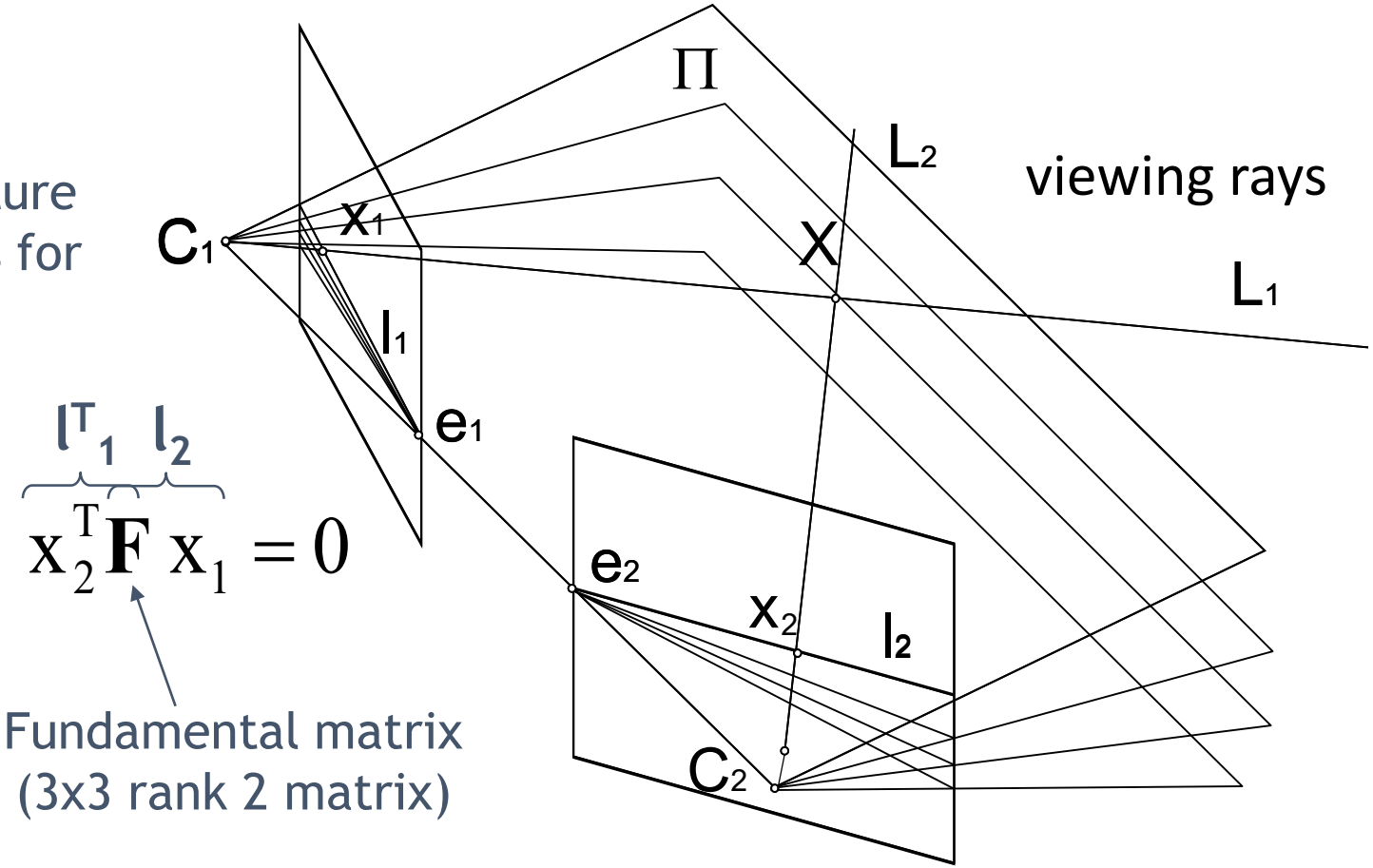
computation of F from pairs of corresponding image points

use $x'_i F x_i = 0$ equations, linear in coeff. F

8 point pairs (linear), 7 point pairs (non-linear), 8+ (least-squares)

Epipolar geometry

Underlying structure in set of matches for rigid scenes



$$\underbrace{l'^{T}_{1}\ l_{2}}\qquad x_2^T \mathbf{F}\, x_1 = 0$$

Fundamental matrix (3x3 rank 2 matrix)

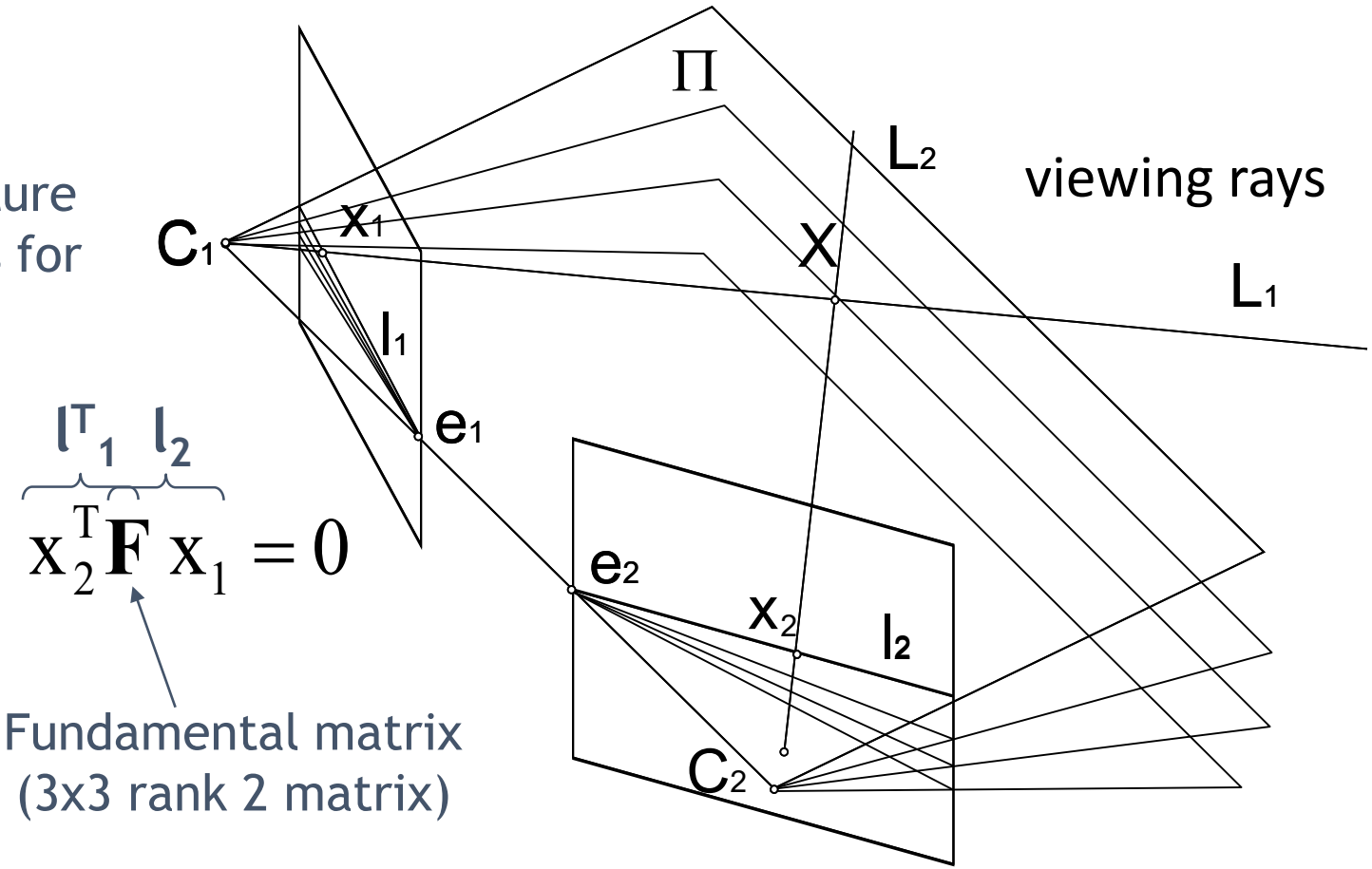
Canonical representation: (see later)

$$P = [I \mid 0] \quad P' = [[e']_\times F + e' v^T \mid \lambda e']$$

1. Computable from corresponding points
2. Simplifies matching
3. Allows to detect wrong matches
4. Related to calibration

Epipolar geometry

Underlying structure in set of matches for rigid scenes



$$\underbrace{l'^{T}_{1}\ l_{2}}\ \underbrace{x_2^{T} F\ x_1}_{} = 0$$

Fundamental matrix (3x3 rank 2 matrix)

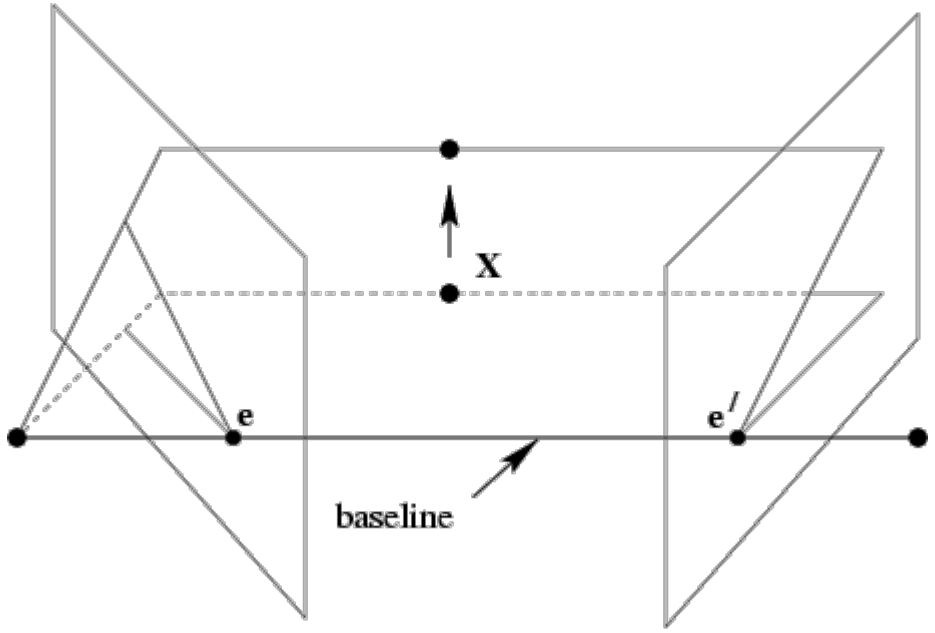
Epipolar planes : 3D cross ratio

Epipolar lines : same 2D cross ratio on both images

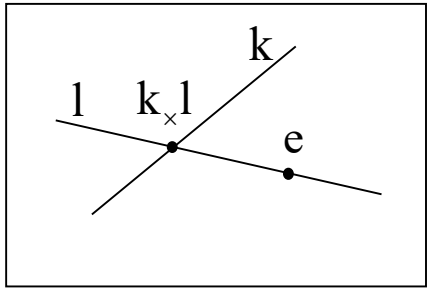
→ projectively related (1D)

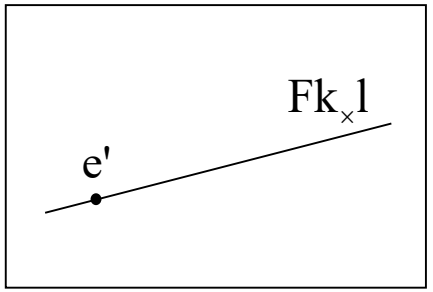
1. Computable from corresponding points
2. Simplifies matching
3. Allows to detect wrong matches
4. Related to calibration

epipolar lines are projectively related: algebraic proof



l epipolar lines through e
take a line k not through e : e.g., take k = e



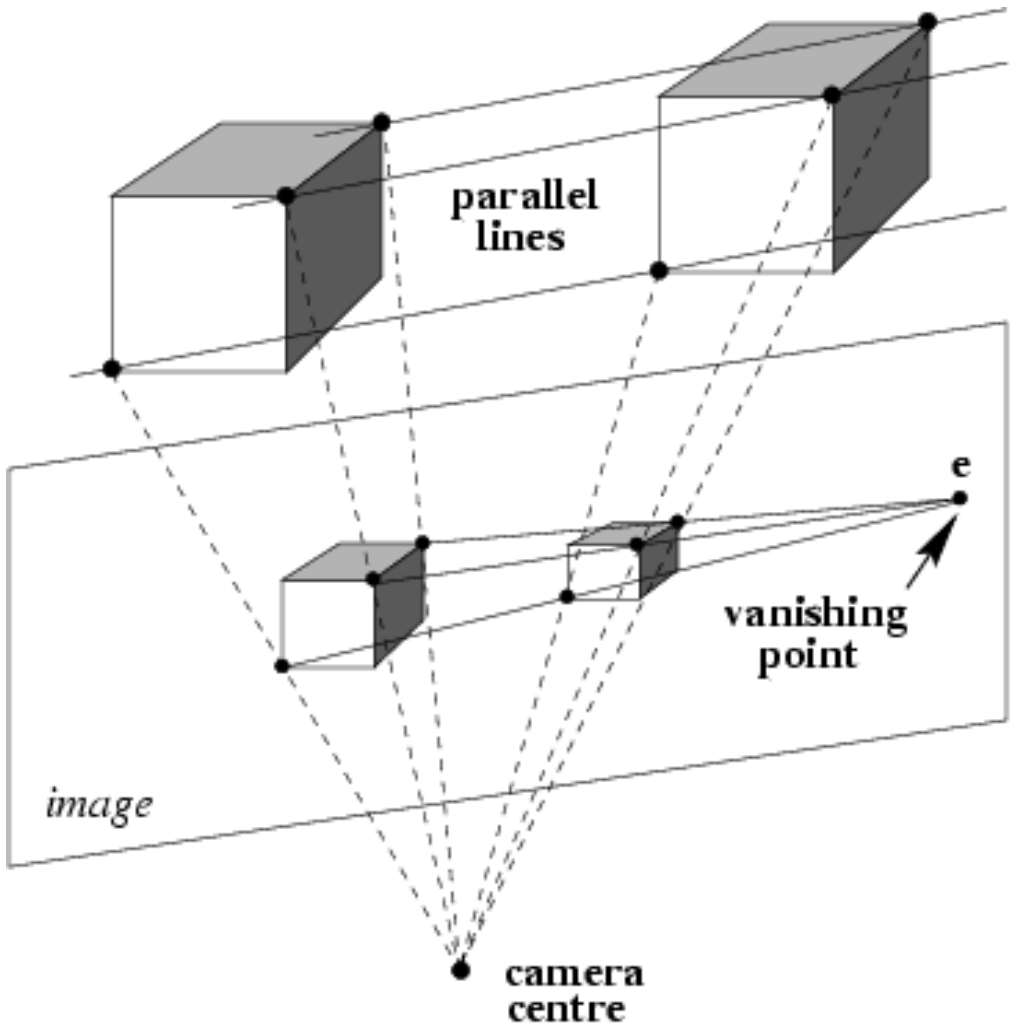


Consider the point $\mathrm{x} = k \cap l = \mathrm{e} \times l = [\mathrm{e}]_\times \, l$

Then $l' = F\mathrm{x} = F[\mathrm{e}]_\times l = Hl$ with $H = F[\mathrm{e}]_\times$

Notice that rank H = 2 < 3. this is OK since
the relation between epipolar lines is 1D projective

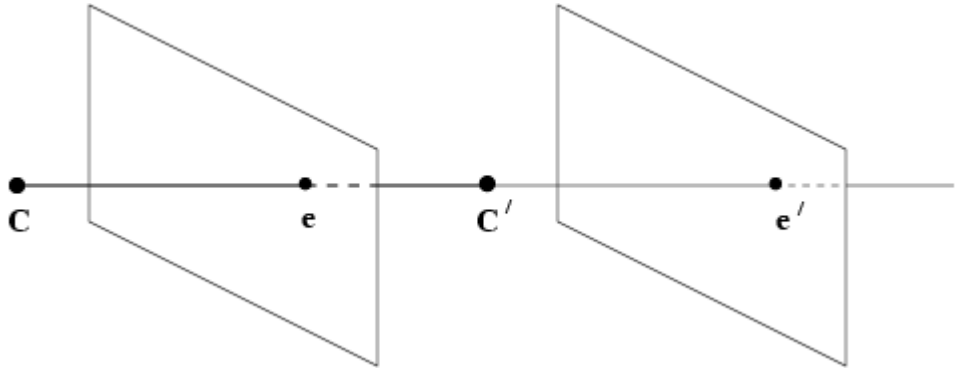
Fundamental matrix for pure translation

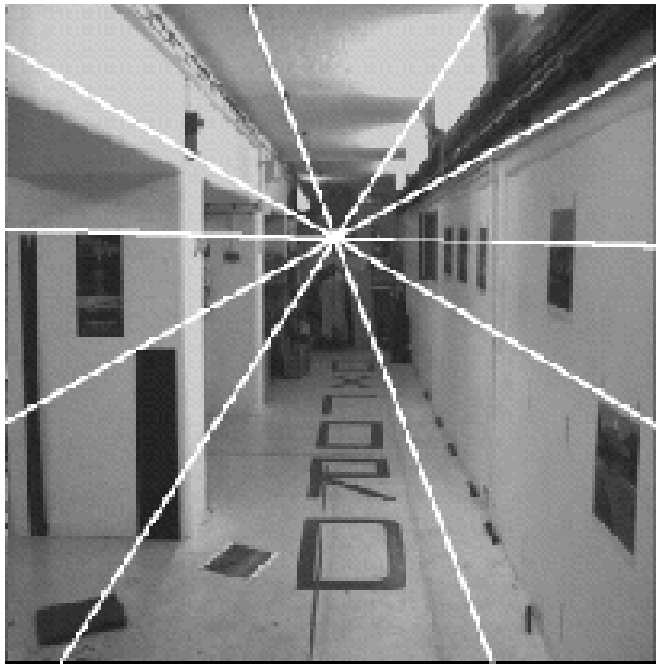


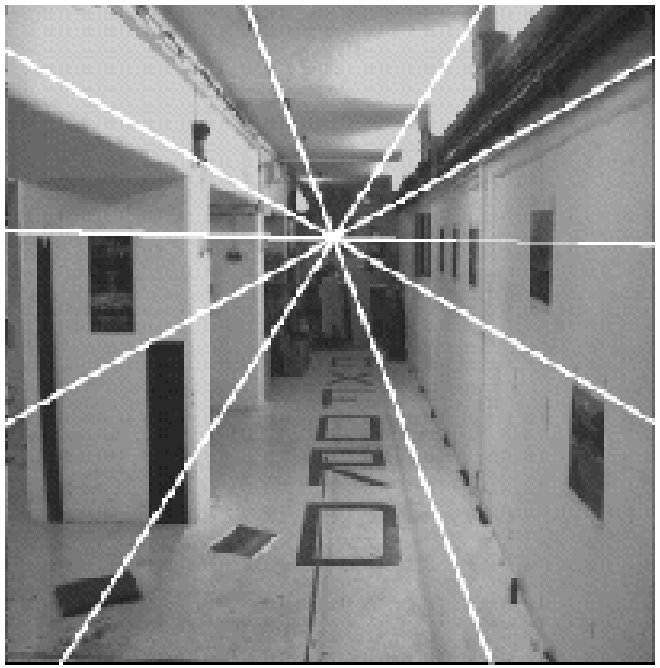
Viewing ray of epipole is parallel to the translation direction

$$\mathrm{K}^{-1}\mathrm{e} = \mathbf{t} \rightarrow \mathrm{e} = \mathrm{K}\mathbf{t}$$

Fundamental matrix for pure translation
the two epipoles are equal







Fundamental matrix for pure translation

$$\mathrm{F} = [\mathrm{e}']_{\times}\mathrm{M}'\mathrm{M}^{-1} = [\mathrm{e}']_{\times}$$

F only 2 d.o.f., $x^T[e]_x x=0 \Rightarrow$ auto-epipolar

example: $\mathbf{e}' = (1,0,0)^{\mathrm{T}} \quad \mathbf{F} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}_{\times} \qquad \mathrm{x}'^{\mathrm{T}}\mathrm{F}\mathrm{x} = 0 \Leftrightarrow y = y'$

$$\mathrm{x} = \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \mathrm{P}\mathrm{X} = \mathrm{K}[\mathrm{I}|0]\mathrm{X} = \mathrm{K}\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} u \\ v \\ Z \end{bmatrix} = Z\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = Z\mathrm{x}_{cart} \rightarrow \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = ZK^{-1}\mathrm{x}_{cart}$$

$$\mathrm{x}' = \mathrm{P}'\mathrm{X} = \mathrm{K}[\mathrm{I}|\mathrm{t}]\begin{bmatrix} ZK^{-1}\mathrm{x}_{cart} \\ 1 \end{bmatrix} = \mathrm{K}[\mathrm{I}|\mathrm{t}]\begin{bmatrix} K^{-1}\mathrm{x}_{cart} \\ 1/Z \end{bmatrix} = \mathrm{x}_{cart} + K\mathrm{t}/Z$$

$$\mathrm{x}' = \mathrm{x}_{cart} + \mathrm{K}\mathrm{t}/Z \text{ this } \mathrm{x}' = \mathrm{x}'_{cart} \text{ if } \mathrm{t}_Z = 0$$

motion starts at x and moves towards (or away from) e, faster depending on *Z*

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see next slide

$$\mathrm{x} = \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \mathrm{PX} = \mathrm{K}[\mathrm{I}|0]\mathrm{X} = \mathbf{K}\begin{bmatrix} \boldsymbol{X} \\ \boldsymbol{Y} \\ \boldsymbol{Z} \end{bmatrix} = \begin{bmatrix} \boldsymbol{u} \\ \boldsymbol{v} \\ \boldsymbol{Z} \end{bmatrix} = Z\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = Z\mathrm{x}_{cart} \rightarrow \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = ZK^{-1}\mathrm{x}_{cart}$$

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$$\mathrm{x}' = \mathrm{x}_{cart} + \mathrm{Kt}/Z \;\text{ this }\; \mathrm{x}' = \mathrm{x}'_{cart} \text{ if } \mathrm{t}_z = 0$$

motion starts at x and moves towards (or away from) e, faster depending on *Z*

$$\text{why is } \mathrm{K}\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} u \\ v \\ Z \end{bmatrix}?$$

$$\mathrm{K}\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} f_x & (s) & U_o \\ 0 & f_y & V_o \\ \mathbf{0} & \mathbf{0} & \mathbf{1} \end{bmatrix}\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} * \\ * \\ Z \end{bmatrix}$$

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In $\begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$, x and y are the cartesian coordinates of the image point x, therefore we indicate

$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \text{ as } \mathrm{x}_{cart}$$

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Yet another expression of P

$$P = [\mathrm{M} \quad \mathrm{m}] \quad O = \begin{bmatrix} o \\ 1 \end{bmatrix} = \mathrm{RNS(P)}$$

$$\mathrm{M} = \mathrm{KR}_{cam \to world} \quad o = t_{world \to cam}$$

From

$$PO = [\mathrm{M} \quad \mathrm{m}] \begin{bmatrix} t_{world \to cam} \\ 1 \end{bmatrix} = \mathrm{M}t_{world \to cam} + \mathrm{m} = 0$$

is $m = -\mathrm{M}t_{world \to cam} = -\mathrm{KR}_{cam \to world} t_{world \to cam}$

But since

$$t_{cam \to world} = -\mathrm{R}t_{world \to cam},$$

Then $m = Kt_{cam \to world}$

Hence $P = [\mathrm{M} \quad \mathrm{m}] = [K\mathrm{R}_{cam \to world} \quad Kt_{cam \to world}]$

$$\boxed{P = K[R \quad t]}$$

where $R \stackrel{\text{def}}{=} \mathrm{R}_{cam \to world}$ and $t \stackrel{\text{def}}{=} t_{cam \to world}$

Fundamental matrix for pure translation

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$\mathrm{x}' = \mathrm{x}_{cart} + \mathrm{K}\mathrm{t}/Z$ this x′ is cartesian, i.e., $\mathrm{x}' = \mathrm{x}'_{cart}$, if $\mathrm{t}_z = 0$ see next slide

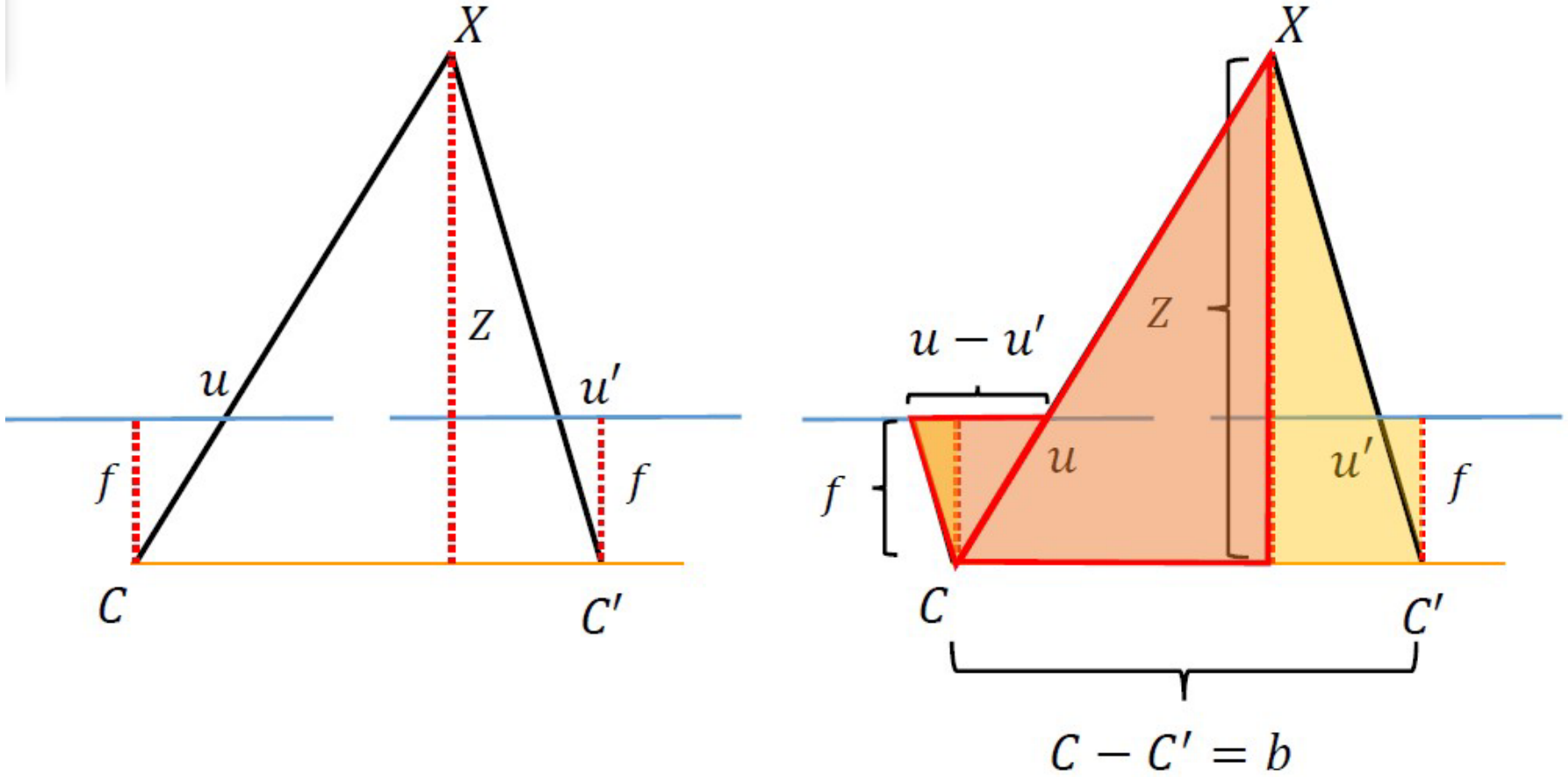
motion starts at x and moves towards (or away from) e, faster depending on *Z*

$$\mathrm{K}\mathbf{t}/\mathrm{Z} = \mathrm{K}\begin{bmatrix} t_X \\ t_Y \\ t_Z \end{bmatrix}/Z == \begin{bmatrix} f_x & (s) & U_o \\ 0 & f_y & V_o \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} t_X \\ t_Y \\ t_Z \end{bmatrix}/Z = \begin{bmatrix} \cdot\cdot \\ \cdot\cdot \\ t_Z/Z \end{bmatrix}$$

$$\text{if } t_Z = 0,$$

$$\mathrm{x}' = \mathrm{x}_{cart} + \mathrm{K}\mathbf{t}/Z = \mathrm{x}_{cart} + \begin{bmatrix} * \\ * \\ 0 \end{bmatrix} = \begin{bmatrix} \cdot \\ \cdot \\ 1 \end{bmatrix} + \begin{bmatrix} \cdot\cdot \\ \cdot\cdot \\ 0 \end{bmatrix} = \begin{bmatrix} * \\ * \\ 1 \end{bmatrix} = \mathrm{x}'_{cart}$$

For equal cameras with coplanar image planes ($t_z = 0$) disparity is inversely proportional to the depth



similar triangles → $\frac{u-u'}{f} = \frac{b}{Z}$

$$x' = x_{cart} + Kt/Z$$

the image of closer objects move faster than the image of distant objects





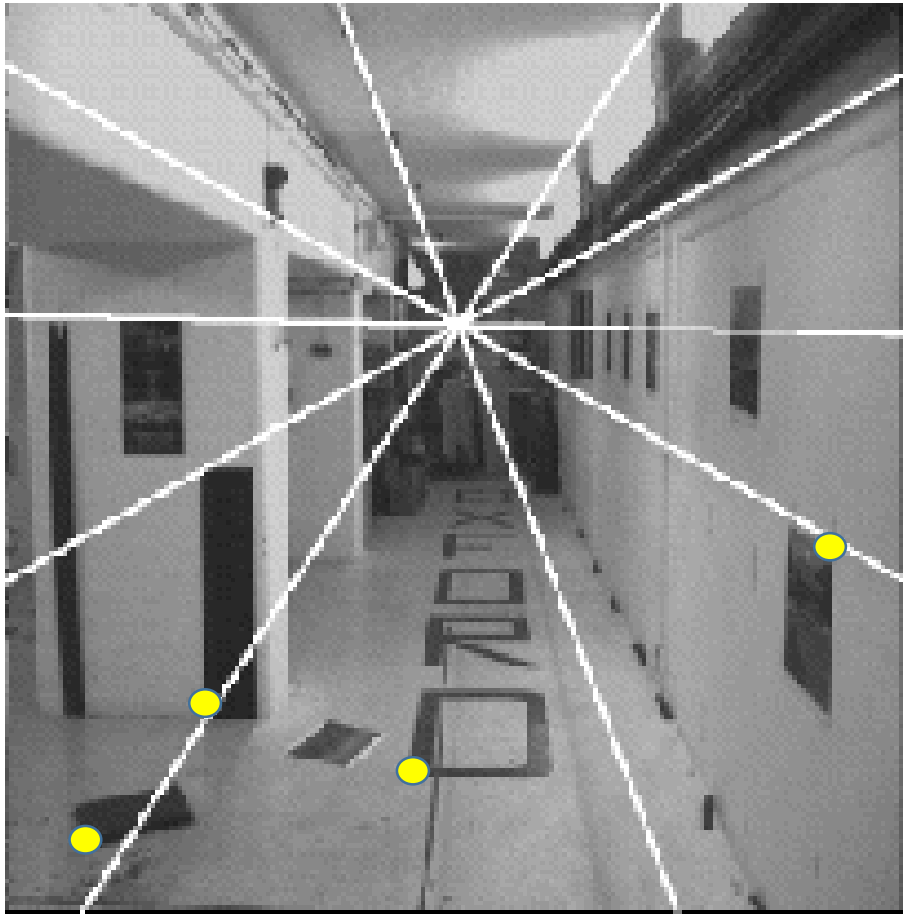
the image of closer objects move faster

than the image of distant objects

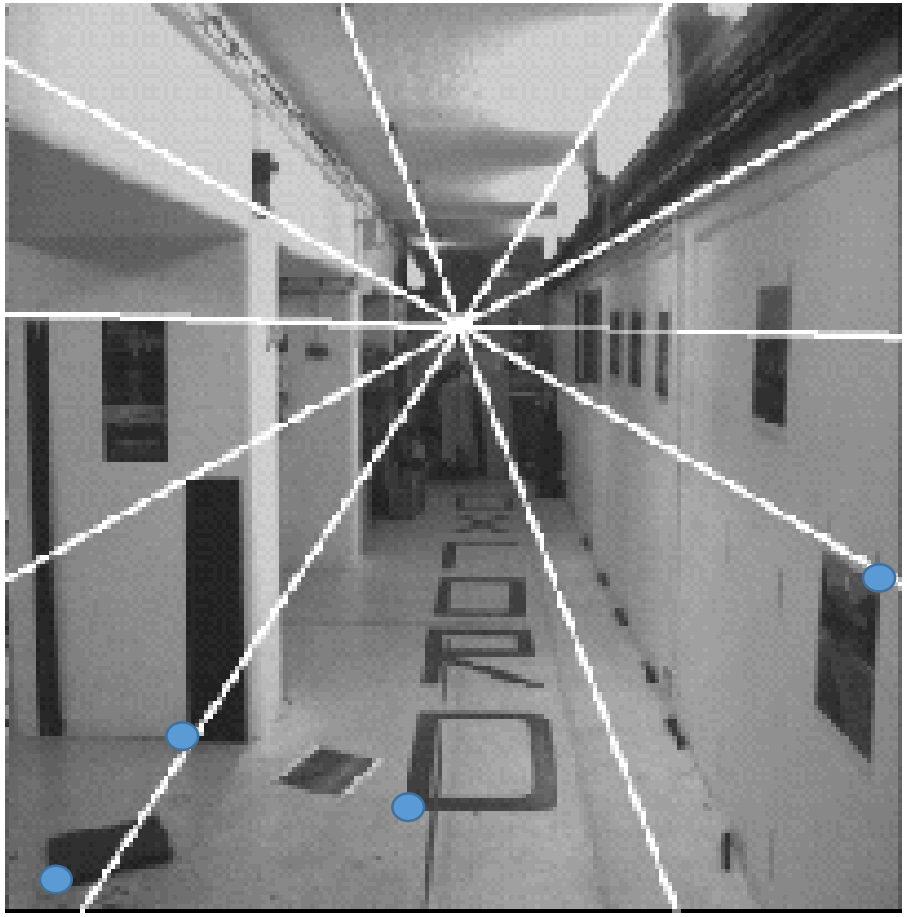


Time to impact

t = 0



t = Δt

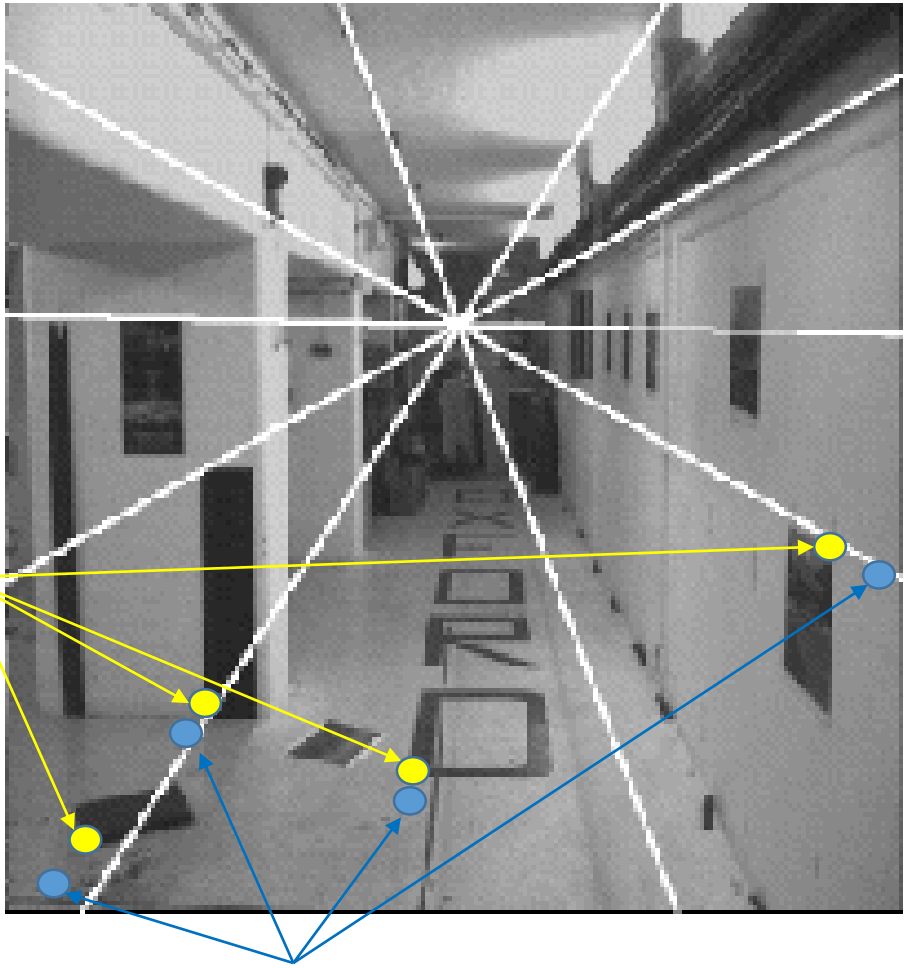


Time to impact

Hyp: constant speed → space proportional to time

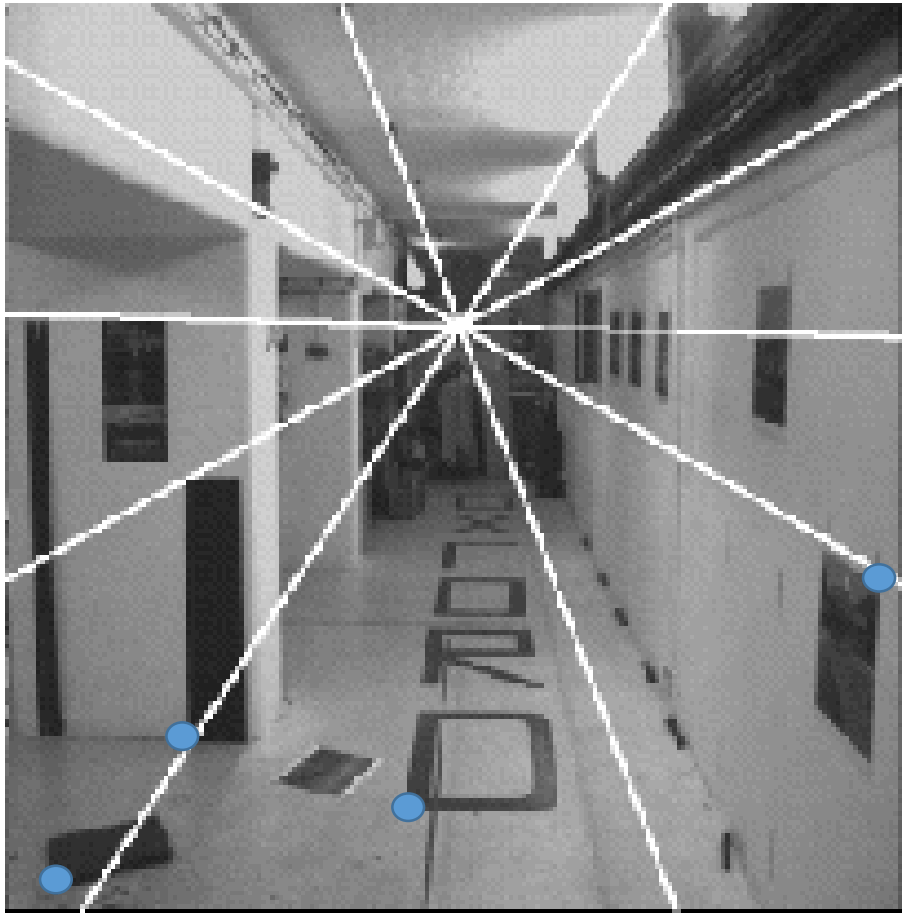
t = 0

t = Δt



t = 0

t = Δt



space proportional to time →

cross ratio of times = cross ratio of distances

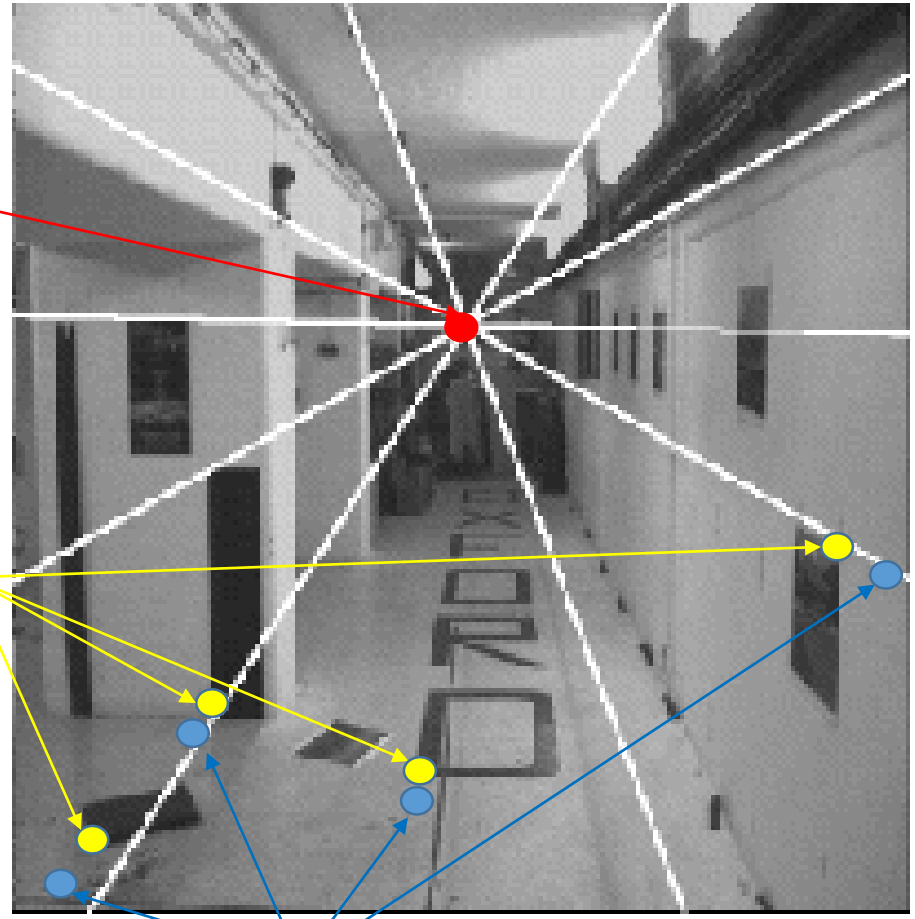
= cross ratio of image coordinates

t = 0

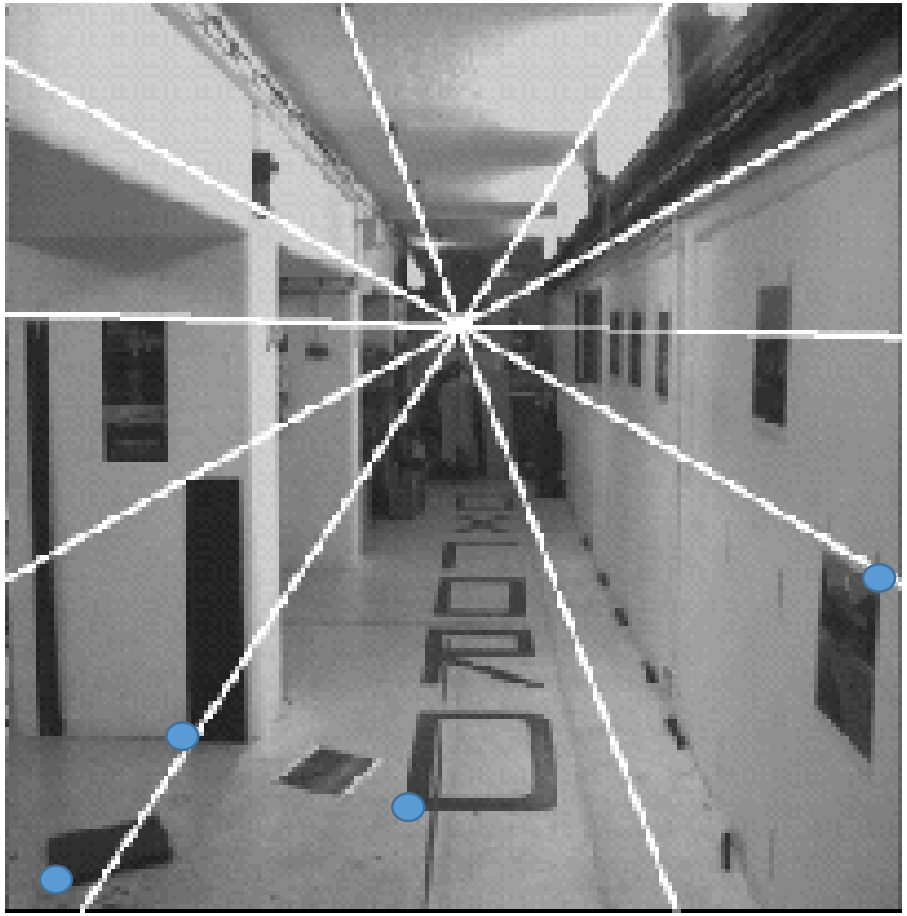
t = Δt

t = ∞

t = 0



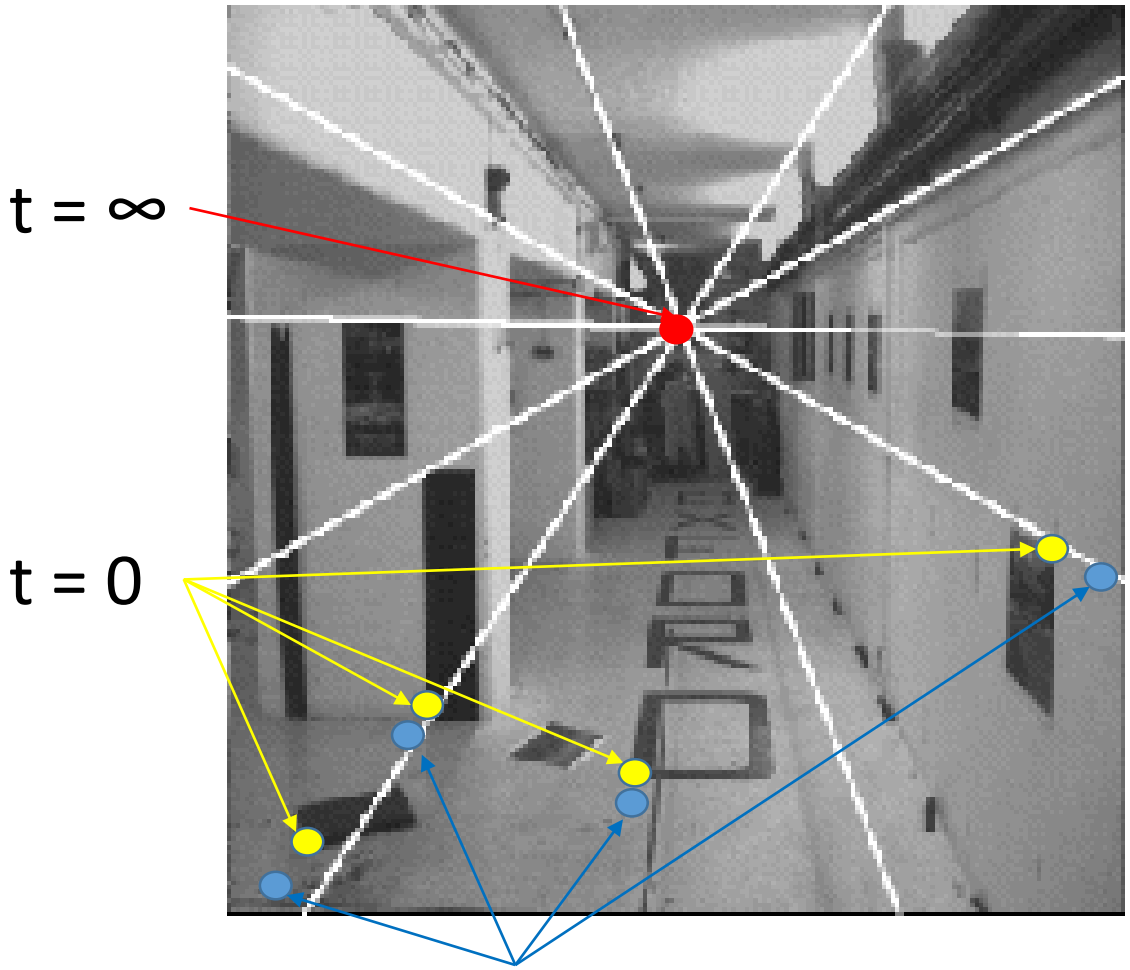
t = Δt

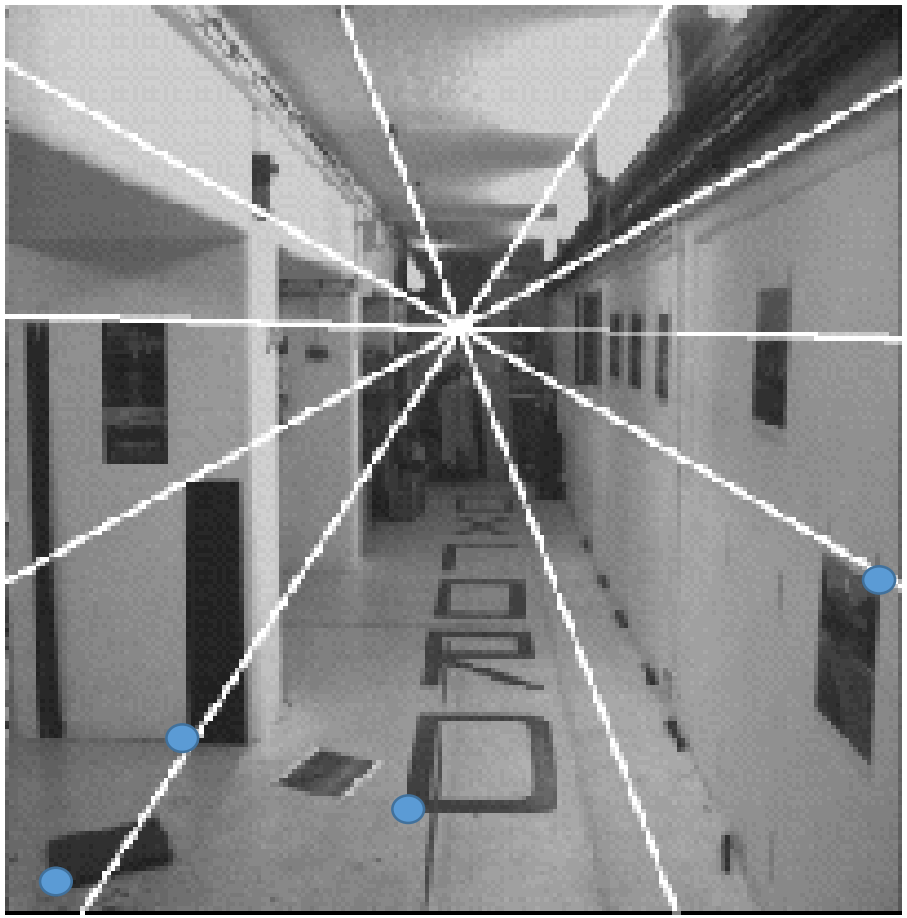


cross ratio of times =
cross ratio of image coordinates
what is the fourth time instant?

t = 0

t = Δt

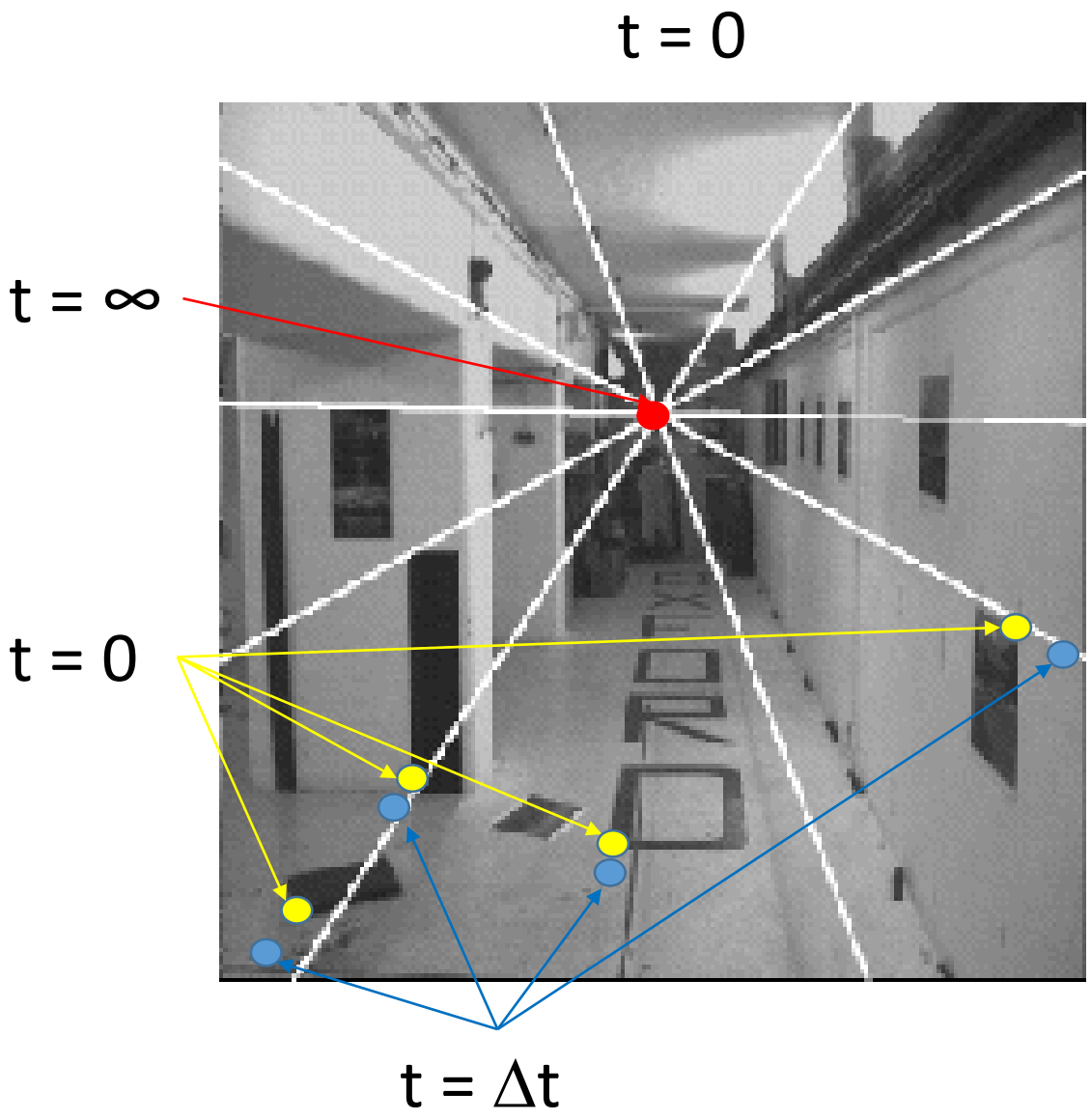


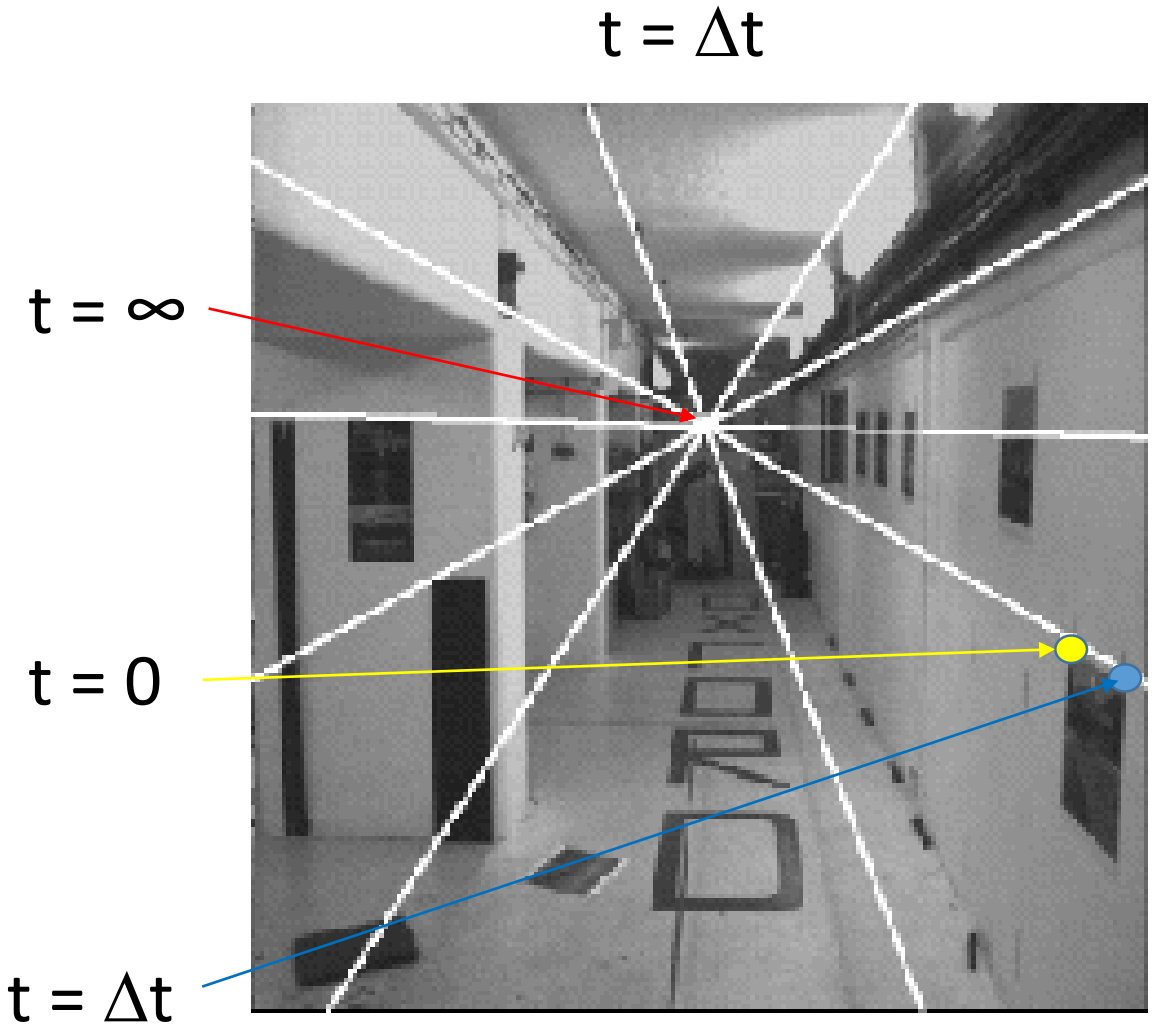


what is the fourth time instant?

time of impact *ti* with lens plane

Each feature has its own impact time instant





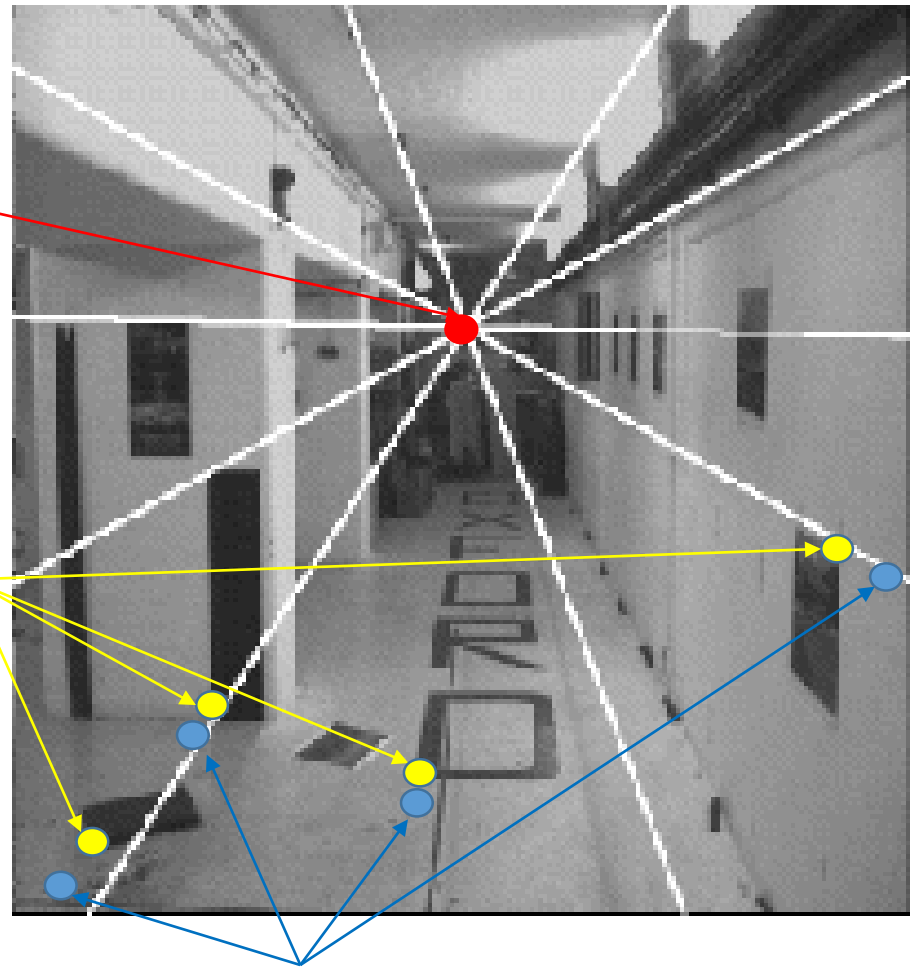
what is the fourth image coordinate?
at time of impact *ti* with lens plane
the image feature is at the ∞

t = 0

t = ∞

t = 0

t = Δt

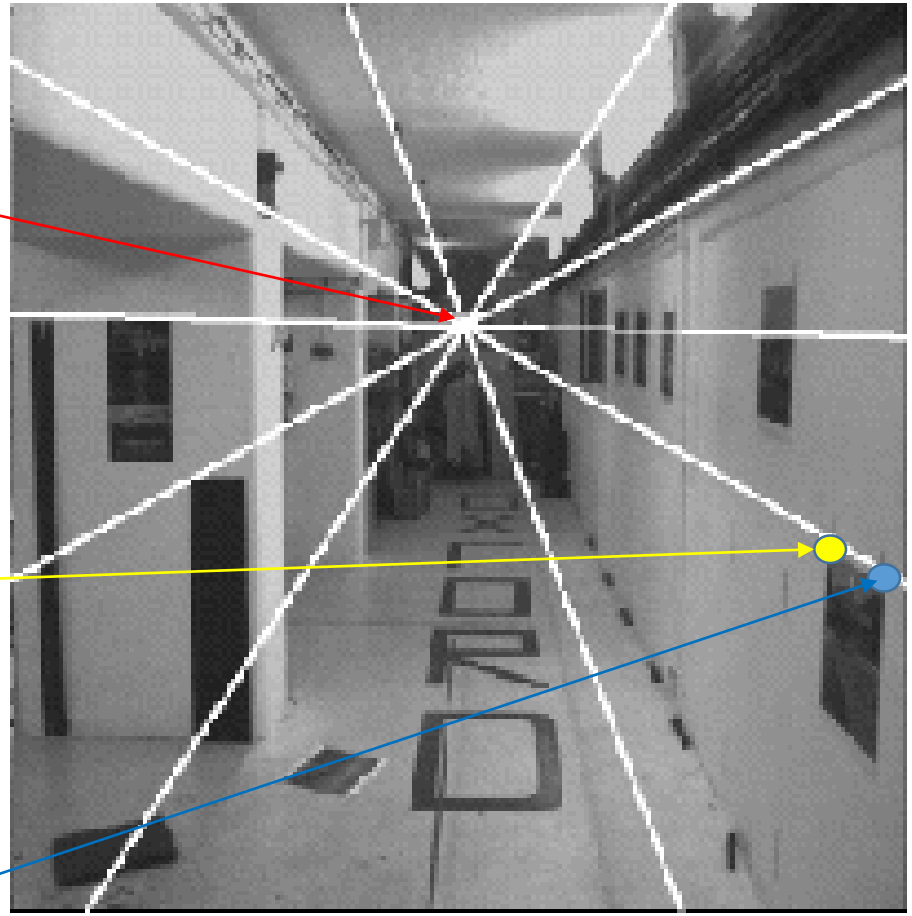


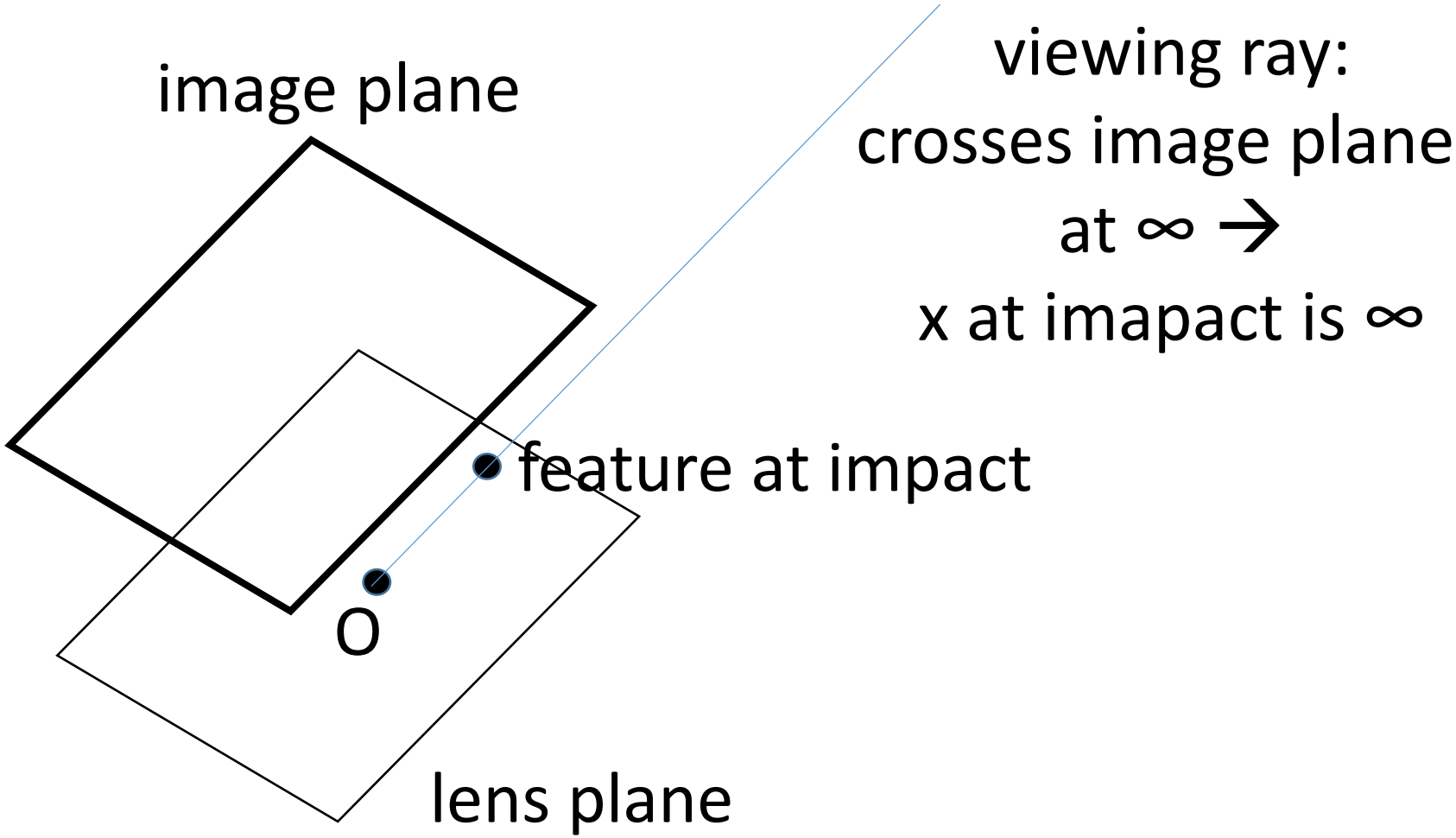
t = Δt

t = ∞

t = 0

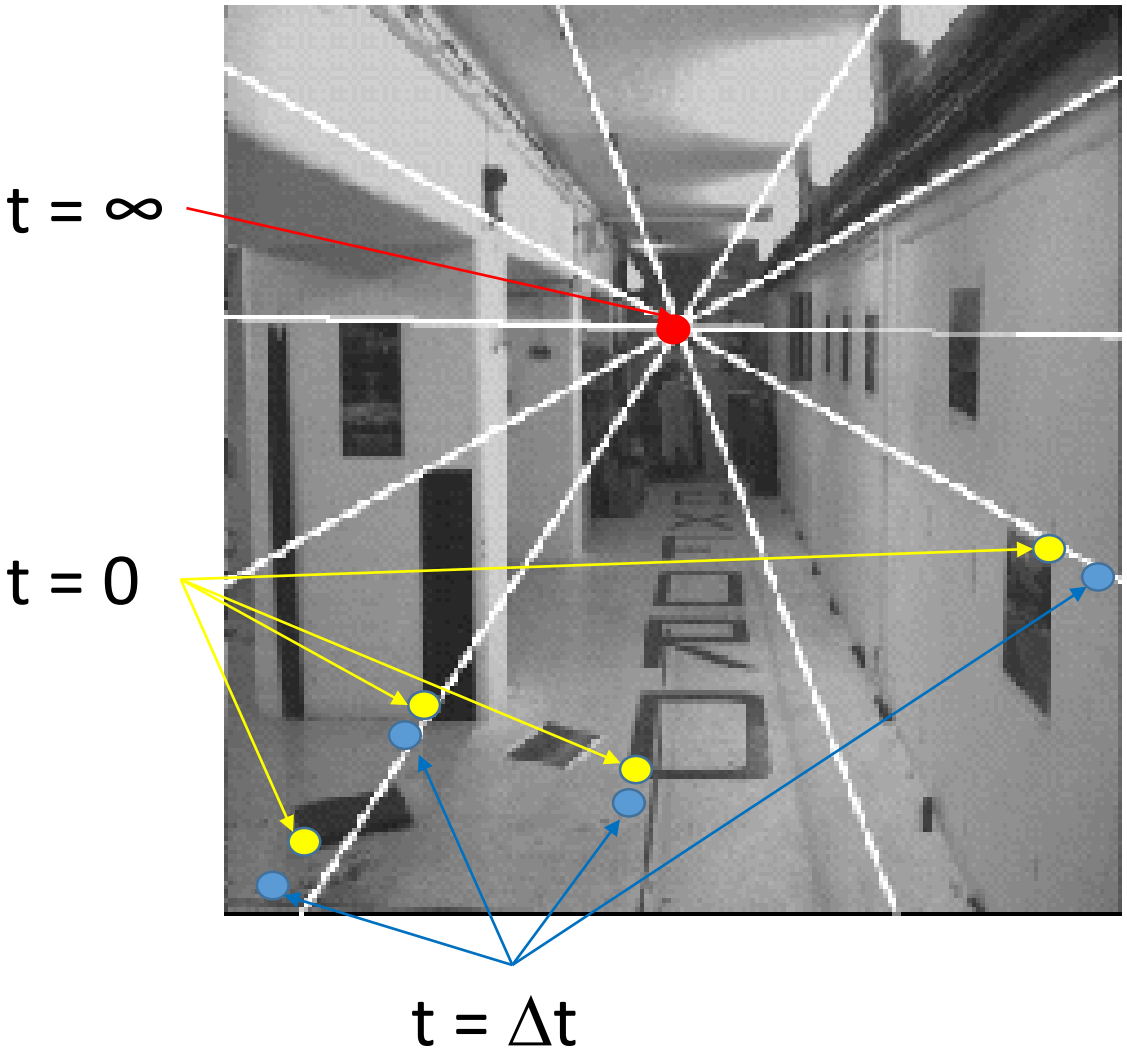
t = Δt

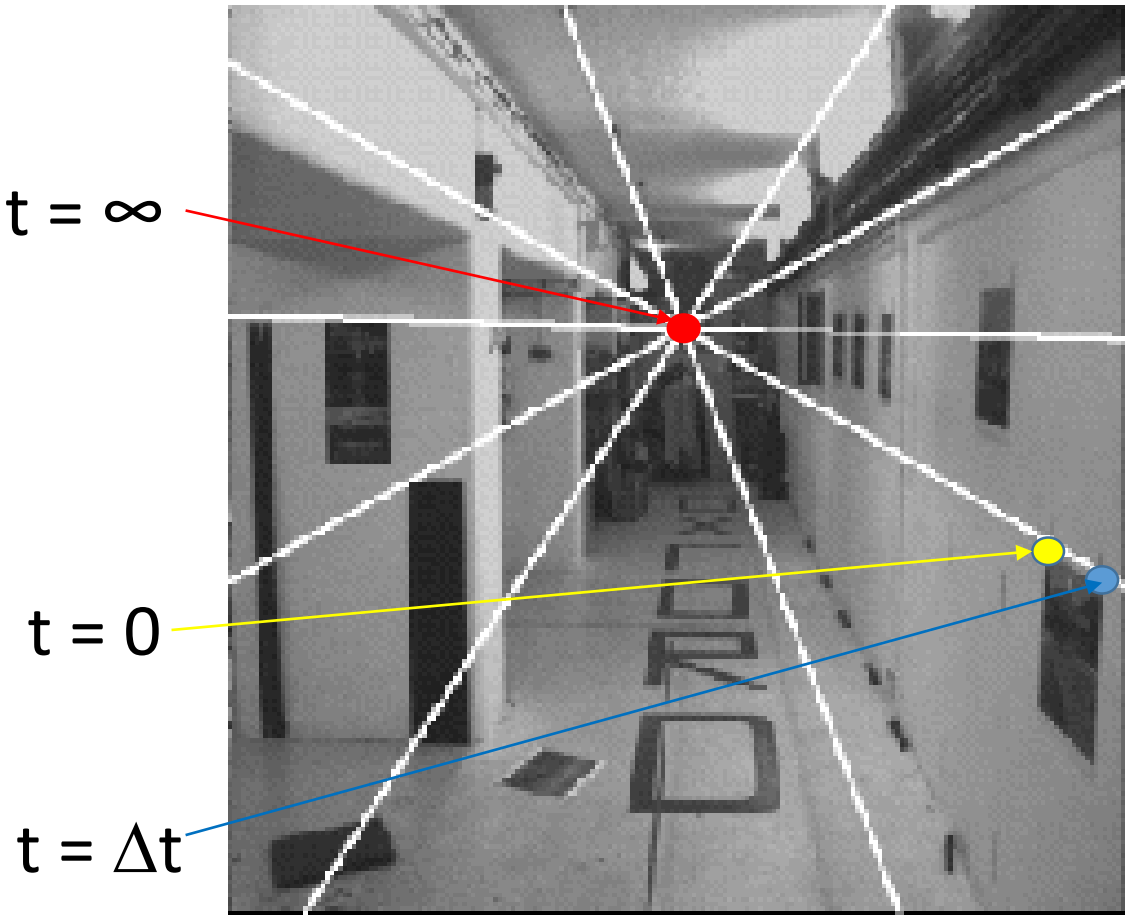



image plane
viewing ray:
crosses image plane
at ∞ →
x at imapact is ∞
feature at impact
O
lens plane

equal cross ratios

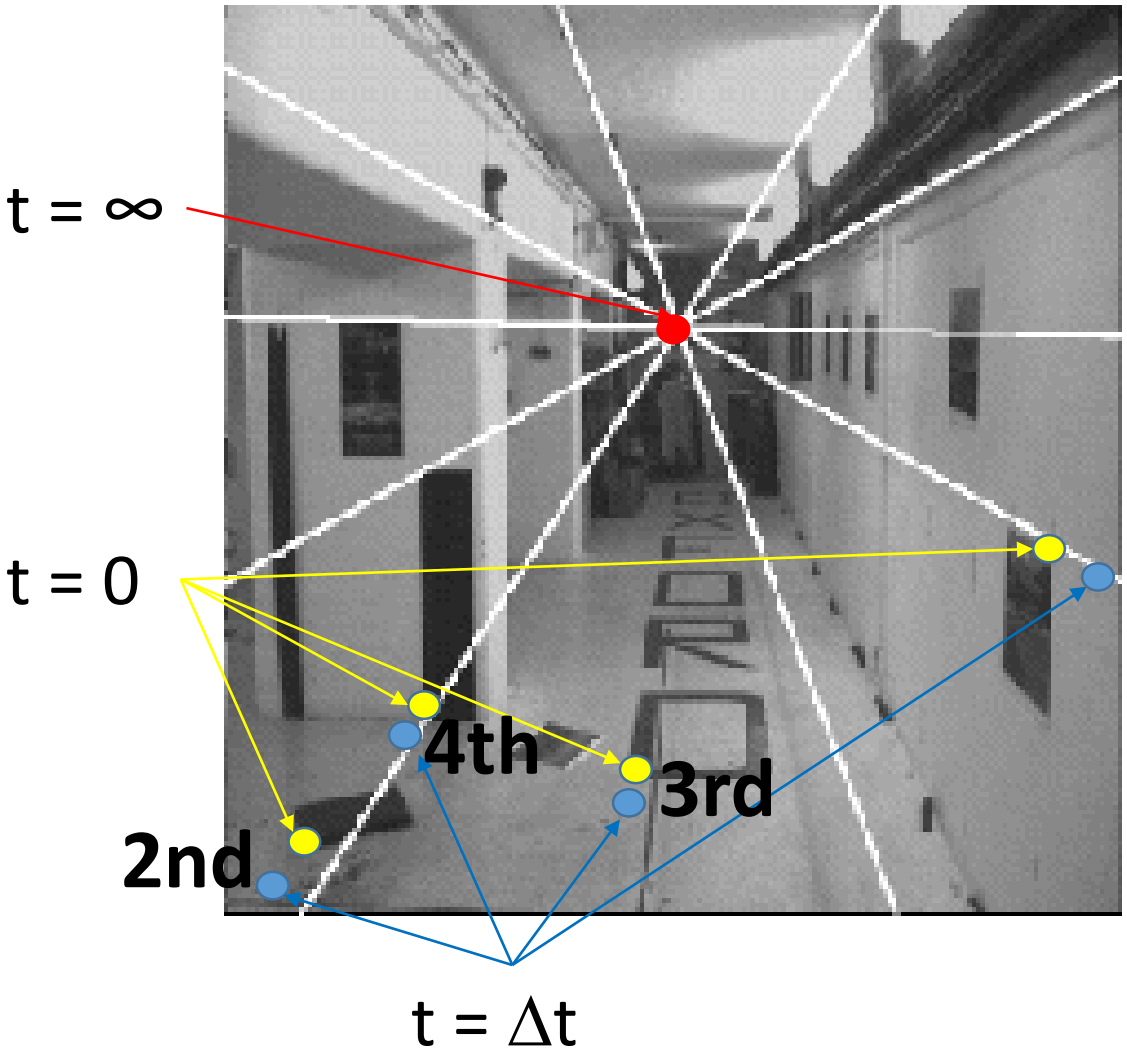
$$\frac{t_i - \Delta t}{t_i - 0} \Big/ \frac{\infty - \Delta t}{\infty - 0} = \frac{\infty - \mathbf{x}'}{\infty - \mathbf{x}} \Big/ \frac{\mathbf{x}_\infty - \mathbf{x}'}{\mathbf{x}_\infty - \mathbf{x}}$$





equal cross ratios

$$t_i = \Delta t \frac{\mathbf{x}_\infty - \mathbf{x}'}{\mathbf{x} - \mathbf{x}'}$$



sort features by increasing time to impact

Translational motion

time to impact of a feature
proportional to the **depth** of the feature
(i.e. distance from lens plane along motion direction)

→

it allows a partial 3D reconstruction

in the case of **calibrated** images (K known)
it allows a **full** (euclidean) 3D reconstruction,
since, for each moving feature, we know both the depth (from its time-to-impact) and the direction (from its viewing ray)

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F only 2 d.o.f., $x^T[e]_x x=0 \Rightarrow$ auto-epipolar

example: $\mathbf{e}' = (1,0,0)^{\mathrm{T}} \quad \mathbf{F} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}_{\times} \quad \mathbf{x}'^{\mathrm{T}}\mathbf{F}\mathbf{x} = 0 \Leftrightarrow y = y'$

$$\mathrm{x} = \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \mathrm{PX} = \mathrm{K}[\mathrm{I}|0]\mathrm{X} = \mathrm{K}\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} u \\ v \\ Z \end{bmatrix} = Z\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = Z\mathrm{x}_{cart} \rightarrow \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = Z\mathrm{K}^{-1}\mathrm{x}_{cart}$$

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$\mathrm{x}' = \mathrm{x}_{cart} + \mathrm{K}\mathrm{t}/Z$ and if $\mathrm{t}_z = 0$ then $\mathrm{x}' = \mathrm{x}'_{cart}$ it is cartesian

motion starts at x and moves towards (or away from) e, faster depending on *Z*

By the way $\mathrm{x}' = \mathrm{x}_{cart} + \mathrm{Kt}/Z$ is a multiple of x'_{cart}

$$\mathrm{x}' = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} + \mathrm{Kt}/Z = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} + \begin{bmatrix} f_x & (s) & U_o \\ 0 & f_y & V_o \\ \mathbf{0} & \mathbf{0} & \mathbf{1} \end{bmatrix} \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix} /Z = \begin{bmatrix} * \\ * \\ 1 + t_z/Z \end{bmatrix}$$

$$\rightarrow$$

$$\mathrm{x}'_{cart} = \frac{\mathrm{x}'}{1+t_z/Z} \rightarrow \mathrm{x}'_{cart}(1 + t_z/Z) = \mathrm{x}' = \mathrm{x}_{cart} + \mathrm{K\,t}/Z$$

but $\mathrm{Kt}/t_z = \mathrm{x}_\infty$ is the common epipole e of the two images, hence

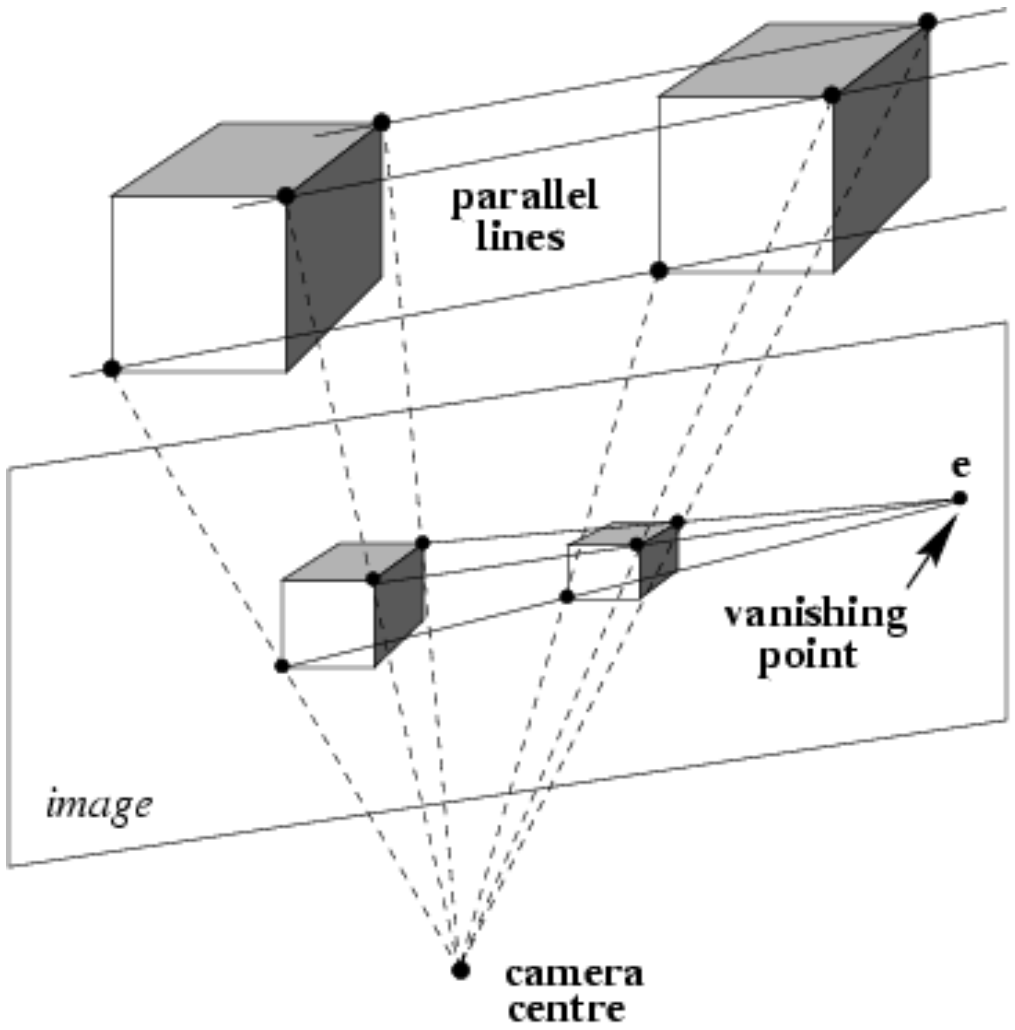
see next slide $\mathrm{x}'_{cart} + \mathrm{x}'_{cart}(t_z/Z) = \mathrm{x}_{cart} + \mathrm{x}_\infty(t_z/Z)$

$$\rightarrow$$

$$Z = t_z \frac{\mathrm{x}_\infty - \mathrm{x}'_{cart}}{\mathrm{x}'_{cart} - \mathrm{x}_{cart}}$$

(ratio of two colinear vectors)

Fundamental matrix for pure translation



Viewing ray of epipole is parallel to the translation direction

$$\mathrm{K}^{-1}\mathrm{e} = \mathbf{t} \rightarrow \mathrm{e} = \mathrm{K}\mathbf{t}$$

By the way $\mathrm{x}' = \mathrm{x}_{cart} + \mathrm{Kt}/Z$ is a multiple of x'_{cart}

$$\mathrm{x}' = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} + \mathrm{Kt}/Z = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} + \begin{bmatrix} f_x & (s) & U_o \\ 0 & f_y & V_o \\ \mathbf{0} & \mathbf{0} & \mathbf{1} \end{bmatrix} \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix} /Z = \begin{bmatrix} * \\ * \\ 1 + t_z/Z \end{bmatrix}$$

→

$$\mathrm{x}'_{cart} = \frac{\mathrm{x}'}{1+t_z/Z} \rightarrow \mathrm{x}'_{cart}(1 + t_z/Z) = \mathrm{x}' = \mathrm{x}_{cart} + \mathrm{K\,t}/Z$$

but $\mathrm{Kt}/t_z = \mathrm{x}_\infty$ is the common epipole e of the two images, hence

$$\mathrm{x}'_{cart} + \mathrm{x}'_{cart}(t_z/Z) = \mathrm{x}_{cart} + \mathrm{x}_\infty(t_z/Z)$$

→

$$Z = t_z \frac{\mathrm{x}_\infty - \mathrm{x}'_{cart}}{\mathrm{x}'_{cart} - \mathrm{x}_{cart}}$$

(ratio of two colinear vectors)

By the way $\mathrm{x}' = \mathrm{x}_{cart} + \mathrm{Kt}/Z$ is a multiple of x'_{cart}

$$\mathrm{x}' = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} + \mathrm{Kt}/Z = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} + \begin{bmatrix} f_x & (s) & U_o \\ 0 & f_y & V_o \\ \mathbf{0} & \mathbf{0} & \mathbf{1} \end{bmatrix} \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix} /Z = \begin{bmatrix} * \\ * \\ 1 + t_z/Z \end{bmatrix}$$

$$\rightarrow$$

$$\mathrm{x}'_{cart} = \frac{\mathrm{x}'}{1+t_z/Z} \rightarrow \mathrm{x}'_{cart}(1 + t_z/Z) = \mathrm{x}' = \mathrm{x}_{cart} + \mathrm{Kt}/Z$$

but $\mathrm{Kt}/t_z = \mathrm{x}_\infty$ is the common epipole of the two images, hence

$$\mathrm{x}'_{cart} + \mathrm{x}'_{cart}(t_z/Z) = \mathrm{x}_{cart} + \mathrm{x}_\infty(t_z/Z)$$

$$\rightarrow$$

$$Z = t_z \frac{\mathrm{x}_\infty - \mathrm{x}'_{cart}}{\mathrm{x}'_{cart} - \mathrm{x}_{cart}}$$

(ratio of two colinear vectors)

Observation

t_z : camera displacement along Z during Δt

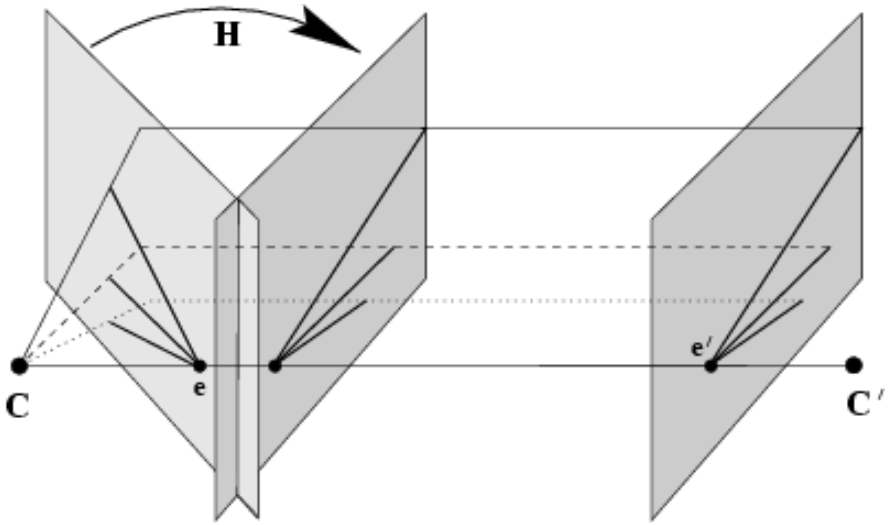
$$Z = t_z \frac{x_\infty - x'_{cart}}{x'_{cart} - x_{cart}}$$

and

$$t_i = \Delta t \frac{x_\infty - x'_{cart}}{x_{cart} - x'_{cart}}$$

are of opposite sign:
this is because the time to impact is positive only if Z decreases with time, which can only occur if the speed $t_z / \Delta t$ is negative

General motion



$$\mathrm{x}'^{\mathsf{T}} \left[\mathrm{e}'\right]_{\times} \hat{\mathrm{x}} = 0$$

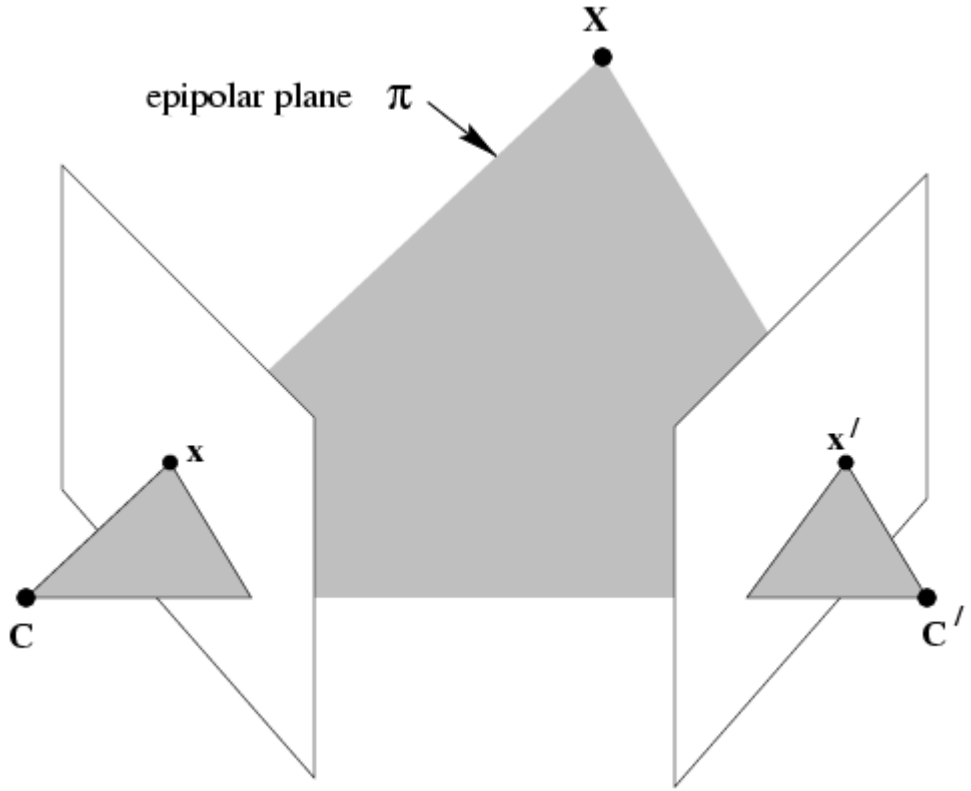
$$\mathrm{x}'^{\mathsf{T}} \left[\mathrm{e}'\right]_{\times} \mathrm{H}\mathrm{x} = 0$$

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = Z\mathrm{K}^{-1}\mathrm{x}_{cart}$$

$$\mathrm{x}' = \mathrm{P}'\mathrm{X} = \mathrm{K}'[\mathrm{R}|\mathrm{t}] \begin{bmatrix} Z\mathrm{K}^{-1}\mathrm{x}_{cart} \\ 1 \end{bmatrix} = \mathrm{K}'[\mathrm{R}|\mathrm{t}] \begin{bmatrix} \mathrm{K}^{-1}\mathrm{x}_{cart} \\ 1 \\ \overline{Z} \end{bmatrix} = \mathrm{K}'\mathrm{R}\mathrm{K}^{-1}\mathrm{x}_{cart} + \mathrm{K}'t/Z$$

$$\mathrm{x}' = \mathrm{K}'\mathrm{R}\mathrm{K}^{-1}\mathrm{x}_{cart} + \mathrm{K}'\mathrm{t}/Z$$

3D reconstruction by triangulation



X = intersection of the viewing rays:

- viewing ray associated to x with camera P
- viewing ray associated to x' with camera P'

Projective ambiguity

Attention: change of notation

$$\mathrm{P}, \mathrm{P}' \rightarrow \mathrm{P}_1, \mathrm{P}_2$$

$$\mathrm{P}_1', \mathrm{P}_2'$$

will be used to indicate modified camera pairs

Remember:

Fundamental matrix of cameras $\mathrm{P}_1 = [\mathrm{M}_1|\mathrm{m}_1], \mathrm{P}_2 = [\mathrm{M}_2|\mathrm{m}_2]$:

$$\mathrm{F} = [\mathrm{e}_2]_\times \mathrm{M}_2 {\mathrm{M}_1}^{-1}$$

$$\mathrm{e}_2 = \mathrm{LNS}(\mathrm{F}) \rightarrow {\mathrm{e}_2}^T \mathrm{F} = 0$$

Projective Ambiguity Theorem

A fundamental matrix F_{12} is compatible with camera pairs (P_1, P_2) and (P_1', P_2')

$$\updownarrow$$

camera pairs are projectively related: i. e. $\exists$ an invertible matrix H_{4x4} such that

$$P_1' = P_1 H^{-1}$$
$$P_2' = P_2 H^{-1}$$

Projective Ambiguity Theorem proof of ↑ direction

camera pairs (P_1, P_2) and (P_1', P_2') have the same fundamental matrix F_{12}

↑

camera pairs are projectively related: i. e. ∃ an invertible matrix H_{4x4} such that

$$P_1' = P_1 H^{-1}$$
$$P_2' = P_2 H^{-1}$$

Projective ambiguity: ↑ proof summary

Suppose that a Fundamental matrix F is compatible with a camera pair (P_1, P_2). Consider a second camera pair (P_1', P_2') such that:

$$P_1' = P_1 H^{-1}$$
$$P_2' = P_2 H^{-1}$$

Triangulating the viewing rays of corresponding pairs of image points $(\mathrm{x}_1, \mathrm{x}_2)$ according to cameras (P_1, P_2), the 3D points X are obtained.

Now apply a projective transformation H to points X : points $X' = HX$ are obtained.

Project the transformed points $X' = HX$ onto cameras (P_1', P_2') :

The **same** image points (→ **same Fundamental matrix** F) are obtained by image-projecting the transformed 3D points $X' = HX$ onto the new cameras (P_1', P_2')

in fact

$$\mathrm{x}'_1 = P_1' X' = P_1 \, H^{-1} X' = P_1 \, H^{-1} HX = P_1 \, X = \mathrm{x}_1$$
$$\mathrm{x}'_2 = P_2' X' = P_2 \, H^{-1} X' = P_2 \, H^{-1} HX = P_2 \, X = \mathrm{x}_2$$

→

given images pairs are compatible not only with true 3D points X

but also with any projectively transformed points $X' = HX$

Projective transformation and invariance

Derivation based purely on projective concepts

$$\hat{x} = Hx, \hat{x}' = H'x' \Rightarrow \hat{F} = H'^{-T} F H^{-1}$$

F invariant to transformations of projective 3D-space

$$x_1 = P_1 X = \left(P_1 H^{-1}\right)(HX) = P_1' X'$$
$$x_2 = P_2 X = \left(P_2 H^{-1}\right)(HX) = P_2' X'$$

$(P_1, P_2) \mapsto F$ unique

$F_{12} \mapsto (P_1, P_2)$ not unique

canonical form

$$P_1 = [I \mid 0]$$
$$P_2 = [M \mid m]$$

$$e_2 = m \rightarrow F_{12} = [m]_\times M$$

useful fact learnt from the ↑ proof

A set of images of a given scene

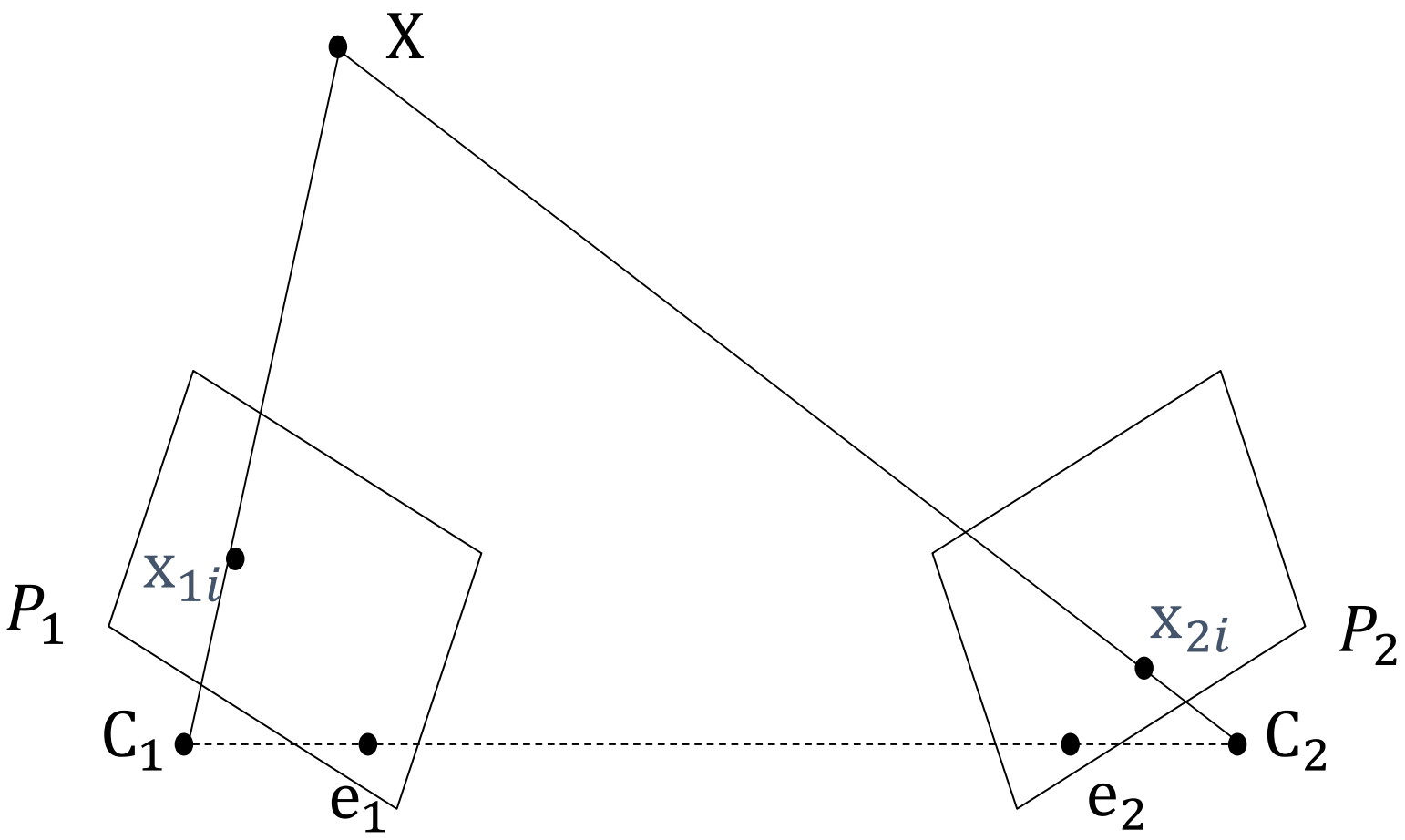
are also compatible with

any projective transformation of the given scene

or

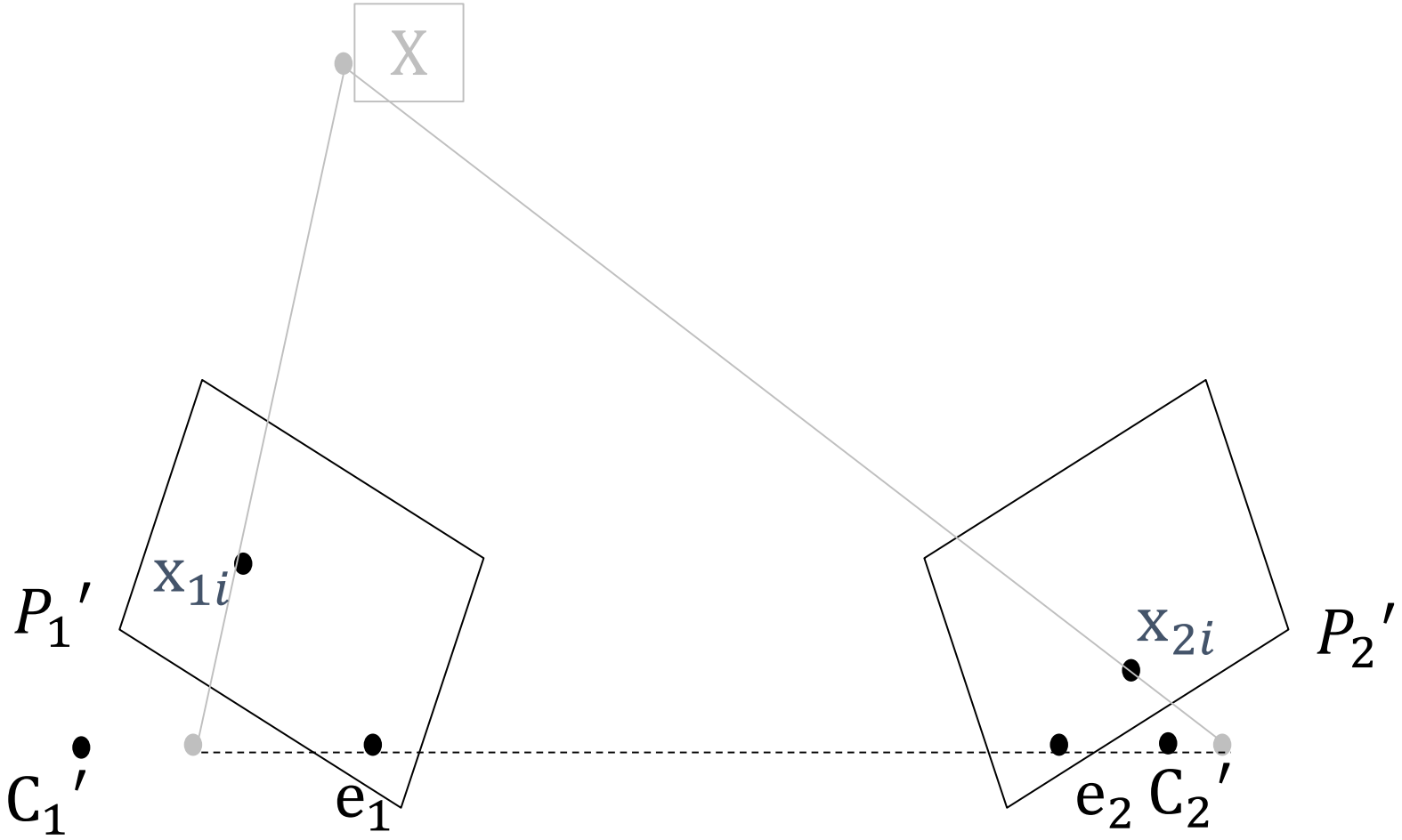
any two projectively related 3D scenes can produce exactly the same images:
by adapting the cameras, i.e. by letting the camera matrices vary

Projective ambiguity F=F (P_1, P_2): given corresponding image points

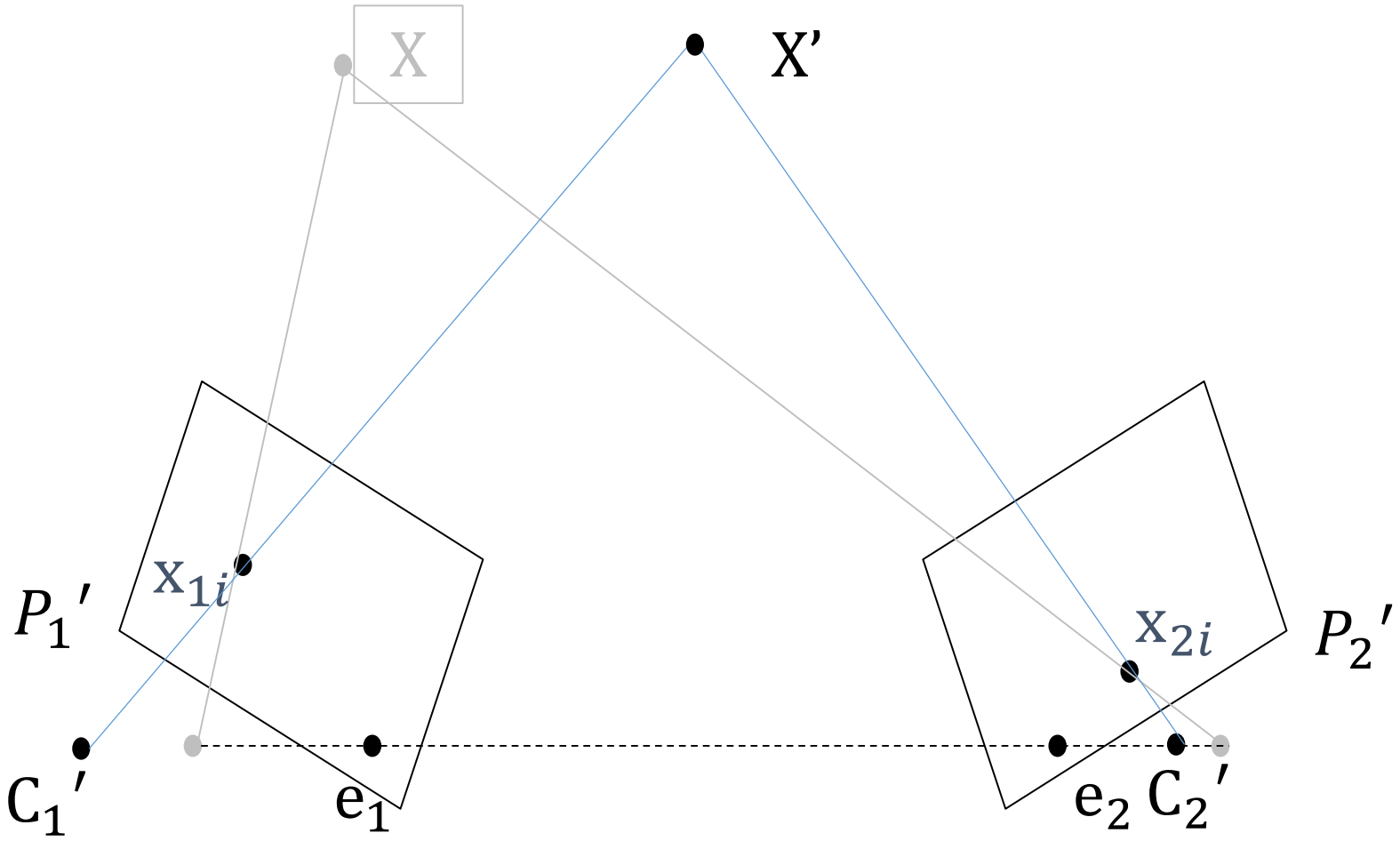


triangulated points X

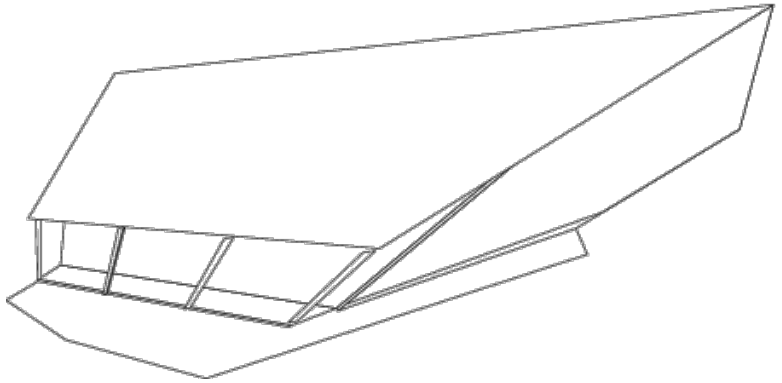
Projective ambiguity F=F (P_1', P_2')=F (P_1, P_2): other cameras with the same F: $P_1' = P_1 \mathrm{H}^{-1}$ $P_2' = P_2 \mathrm{H}^{-1}$



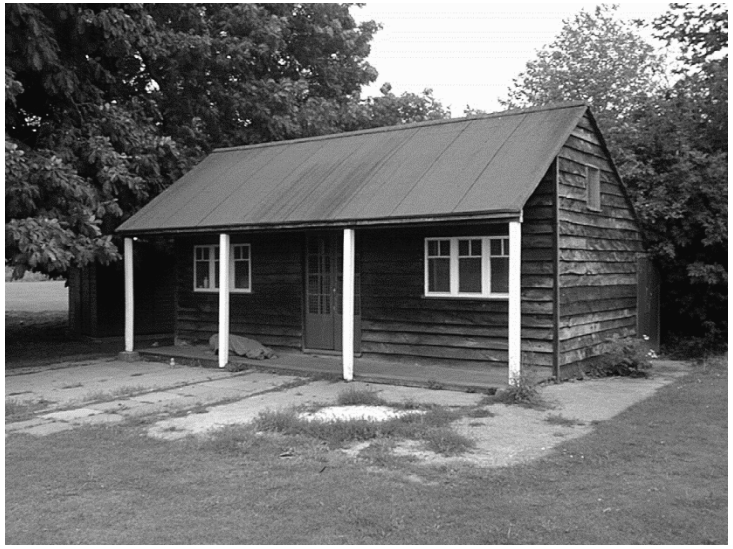
Projective ambiguity F=F (P_1', P_2')=F (P_1, P_2):
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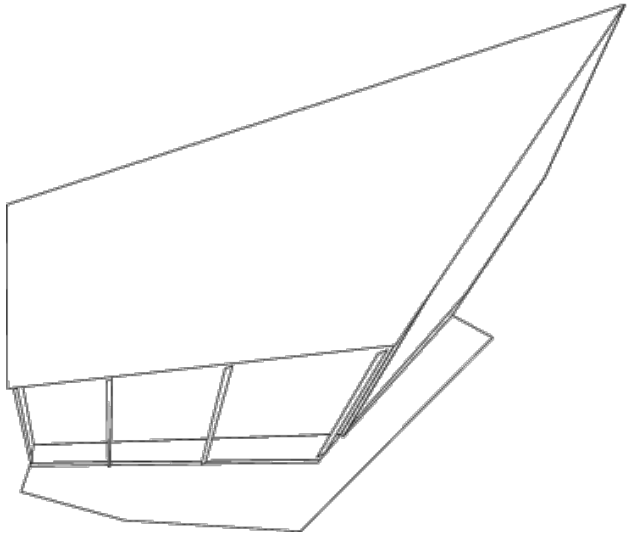
triangulated points $\mathrm{X}' = \mathrm{HX}$



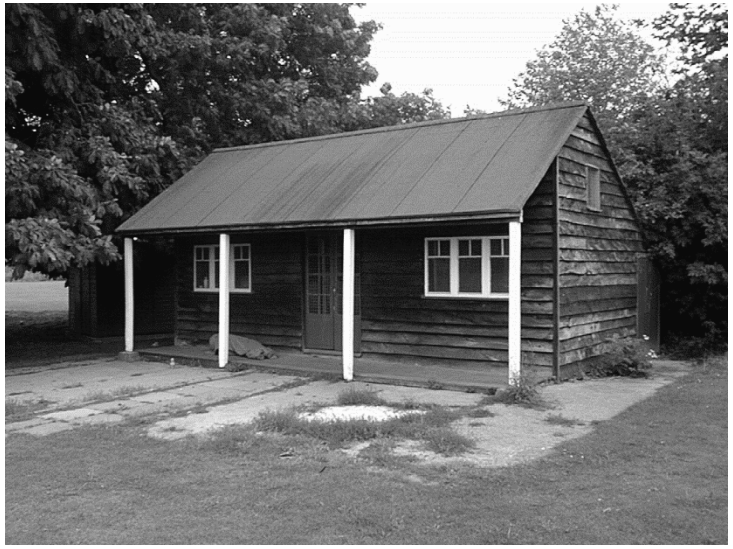
X_i







X_i'





Camera pairs in canonical form

From ↑ part of the theorem

two camera pairs that are projectively related have the same F

→

if camera pair (P_1^o, P_2^o) has F as fundamental matrix, with $P_1^o = [\mathrm{M} \quad \mathrm{m}]$,

then take $\mathrm{H} = \begin{bmatrix} & \mathrm{M} & & \mathrm{m} \\ 0 & 0 & 0 & 1 \end{bmatrix}$: $P_1^o = [\mathrm{M} \quad \mathrm{m}] = [\mathrm{I} \quad 0]\mathrm{H}$

therefore, also $\boldsymbol{P_1} \stackrel{\text{def}}{=} [\mathbf{I} \quad \mathbf{0}] = P_1^o H^{-1}$ and $\boldsymbol{P_2} \stackrel{\text{def}}{=} [\mathbf{A} \quad \mathbf{a}] = P_2^o H^{-1}$

have the same F as fundamental matrix!! Cameras in **canonical form** for F :

$$\boxed{(P_1, P_2) = ([\mathrm{I} \quad 0], [\mathrm{A} \quad \mathrm{a}])}$$

In addition, the epipole of pair $(P_1, P_2) = (([\mathrm{I} \quad 0], [\mathrm{A} \quad \mathrm{a}])$ is $[\mathrm{A} \quad \mathrm{a}] \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \mathrm{a}$

and $F = [\mathrm{e}_2]_\times \mathrm{M}_2 {\mathrm{M}_1}^{-1} = [a]_\times \mathrm{A}\mathrm{I}^{-1} = [a]_\times \mathrm{A}$

image by P_2 of the center of P_1

Projective Ambiguity Theorem: proof of ↓ direction

A fundamental matrix F_{12} is compatible with camera pairs (P_1, P_2) and (P_1', P_2')

↓

camera pairs are projectively related: i. e. ∃ an invertible matrix H_{4x4} such that

$$P_1' = P_1 H^{-1}$$
$$P_2' = P_2 H^{-1}$$

Projective ambiguity of cameras given F

↓ easier proof with cameras in canonical form

if F is the same both for (P_1, P_2) and for (P_1', P_2') ➔

∃ a projective transformation H so that $P_1' = P_1 H^{-1}$ and $P_2' = P_2 H^{-1}$

$$\mathrm{P}_1 = [\mathrm{I}|0]\,;\, \mathrm{P} = \mathrm{P}_2 = [\mathrm{A}|\mathrm{a}] \quad \text{and} \quad \mathrm{P}_1' = [\mathrm{I}|0];\, \mathrm{P}' = \mathrm{P}_2' = [\tilde{\mathrm{A}}|\tilde{\mathrm{a}}]$$

$$\mathrm{F} = [\mathrm{a}]_\times \mathrm{A} = [\tilde{\mathrm{a}}]_\times \tilde{\mathrm{A}}$$

lemma: $\tilde{\mathbf{a}} = k\mathbf{a}$ **and** $\tilde{\mathbf{A}} = k^{-1}(\mathbf{A} + \mathbf{a}\mathbf{v}^{\mathbf{T}})$

proof: $\mathrm{a}^T \mathrm{F} = \mathrm{a}^T [\mathrm{a}]_\times \mathrm{A} = 0$ and similarly $\tilde{\mathrm{a}}\mathrm{F} \Rightarrow 0$ $\tilde{\mathrm{a}} = \mathrm{k}\mathrm{a}$ since they are both LNS of the rank 2 matrix F

$[\mathrm{a}]_\times \mathrm{A} = [\tilde{\mathrm{a}}]_\times \tilde{\mathrm{A}} \Rightarrow [\mathrm{a}]_\times (k\tilde{\mathrm{A}} - \mathrm{A}) = 0 \Rightarrow$ all columns of $(k\tilde{\mathrm{A}} - \mathrm{A})$ are multiples of a $\Rightarrow (k\tilde{\mathrm{A}} - \mathrm{A}) = \mathrm{a}\mathrm{v}^{\mathrm{T}}$

in fact, a $\times (k\tilde{\mathrm{A}} - \mathrm{A}) = 0 \Rightarrow$ a is «parallel» to each column vector in $(k\tilde{\mathrm{A}} - \mathrm{A})$

take $\mathrm{H}^{-1} = \begin{bmatrix} k^{-1}\mathrm{I} & 0 \\ k^{-1}\mathrm{v}^{\mathrm{T}} & k \end{bmatrix}$ $\mathrm{P}_1 \mathrm{H}^{-1} = [I|0] = \mathrm{P}_1'$ and from $(k\tilde{\mathrm{A}} - \mathrm{A}) = \mathrm{a}\mathrm{v}^{\mathrm{T}}$ is $\tilde{\mathrm{A}} = k^{-1}(\mathrm{A} + \mathrm{a}\mathrm{v}^{\mathrm{T}})$

$$\mathrm{P}_2 \mathrm{H}^{-1} = [\mathrm{A}|\mathrm{a}] \begin{bmatrix} k^{-1} I & 0 \\ k^{-1}\mathrm{v}^{\mathrm{T}} & k \end{bmatrix} = \left[k^{-1}(\mathrm{A} + \mathrm{a}\mathrm{v}^{\mathrm{T}}) \,\middle|\, k\mathrm{a} \right] = [\tilde{\mathrm{A}}|\tilde{\mathrm{a}}] = \mathrm{P}_2'$$

(22-15=7, ok)

useful fact learnt from the ↓ proof

if

cameras $\mathrm{P}_1 = [\mathrm{I}|0]$ and $\mathrm{P}_2 = [\mathrm{A}|\mathrm{a}]$ are compatible with a Fundamental matrix F_{12}

then

also cameras $\mathbf{P_1}' = [\boldsymbol{I}|\mathbf{0}]$ and $\mathbf{P_2}' = [\,\mathbf{A}+\mathbf{a}\mathbf{v}^{\boldsymbol{T}}\big|\boldsymbol{\lambda a}]$
are compatible with F_{12} , for any vector v and scalarλ

Most general canonical cameras given F

A possible choice of cameras compatible with a given F:

$$\mathrm{P}_1 = [\mathrm{I}|0] \quad \mathrm{P}_2 = [[\mathrm{e}_2]_\times F \mid \mathrm{e}_2], \quad (\mathrm{e}_2 = \mathrm{LNS}\ F)$$

since the fundamental matrix of $(\mathrm{P}_1, \mathrm{P}_2)$ is F: in fact for canonical cameras is

$$F = [\mathrm{m}]_\times \mathrm{M} = [\mathrm{e}_2]_\times \mathrm{M} = [\mathrm{e}_2]_\times [\mathrm{e}_2]_\times F = -\|\mathbf{e_2}\|^2 \boldsymbol{F} \Leftrightarrow F$$

However, $\mathrm{M} = [\mathrm{e}_2]_\times F$ is singular → camera P_2 is degenerate

Canonical representation:

from theorem proof with $A = [\mathrm{e}_2]_\times F$, $\mathrm{a} = \mathrm{e}_2$, divide by *k* and take $\lambda = 1/k^2$

$$\mathrm{P}_1 = [\mathrm{I}|0] \quad \mathrm{P}_2 = [[\mathrm{e}_2]_\times F + \mathrm{e}_2 \mathrm{v}^\mathrm{T} | \lambda \mathrm{e}_2]$$

is the **most general camera pair** in canonical form, that is compatible with F

Most general canonical cameras given F

A possible choice of cameras compatible with a given F:

$$\mathrm{P}_1 = [\mathrm{I}|0] \quad \mathrm{P}_2 = [[\mathrm{e}_2]_\times F \mid \mathrm{e}_2], \quad (\mathrm{e}_2 = \mathrm{LNS}\ F)$$

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see next slides

However, $\mathrm{M} = [\mathrm{e}_2]_\times F$ is singular → camera P_2 is degenerate

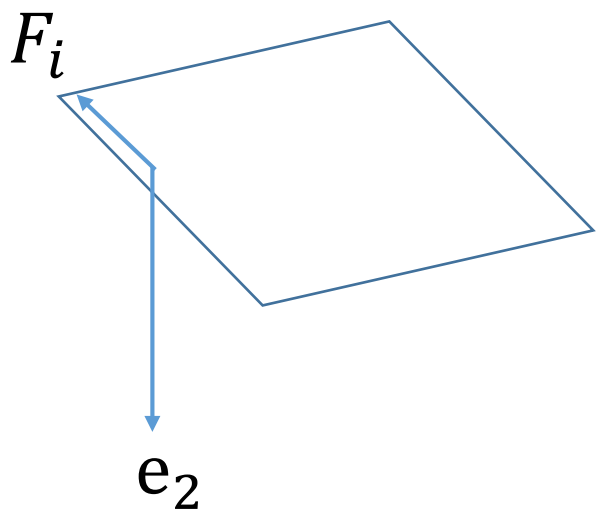
Canonical representation:

from theorem proof with $A = [\mathrm{e}_2]_\times F$, $\mathrm{a} = \mathrm{e}_2$, divide by k and take $\lambda = 1/k^2$

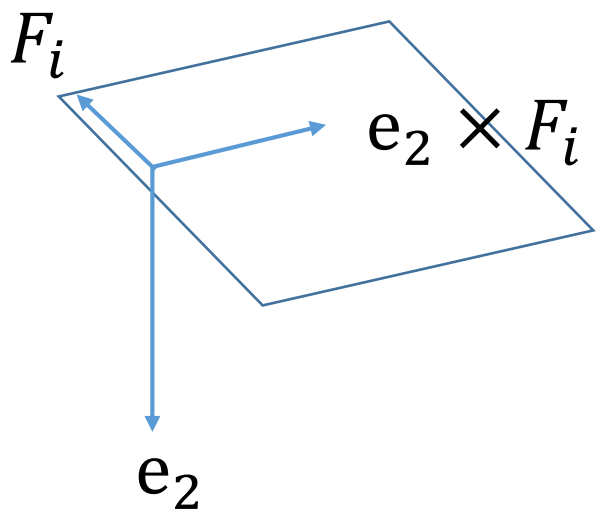
$$\mathrm{P}_1 = [\mathrm{I}|0] \quad \mathrm{P}_2 = [[\mathrm{e}_2]_\times F + \mathrm{e}_2 \mathrm{v}^\mathrm{T} | \lambda \mathrm{e}_2]$$

is the **most general camera pair** in canonical form, that is compatible with F

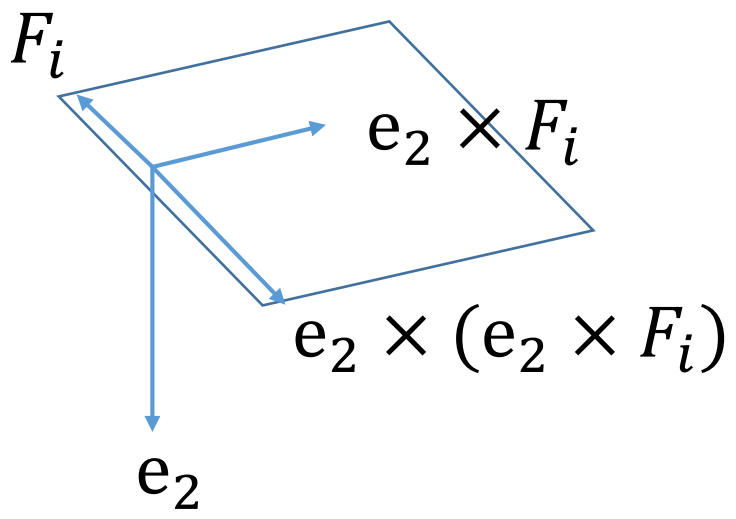
e_2 is LNS() → $e_2^T F = e_2^T [F_1 \quad F_2 \quad F_3] = 0 \rightarrow e_2 \perp F_i$



e_2 is LNS() → $e_2^T F = e_2^T [F_1 \quad F_2 \quad F_3] = 0 \rightarrow e_2 \perp F_i$



e_2 is LNS() → $\mathrm{e}_2^T\,\mathrm{F} = \mathrm{e}_2^T\,[F_1 \quad F_2 \quad F_3] = 0 \rightarrow \mathrm{e}_2 \perp F_i$



$$[\mathrm{e}_2]_\times[\mathrm{e}_2]_\times F_i = -\|\mathrm{e}_2\|^2 F_i$$

$$\boxed{[\mathrm{e}_2]_\times[\mathrm{e}_2]_\times \mathrm{F} = -\|\mathrm{e}_2\|^2 \mathrm{F}}$$

Most general canonical cameras given F

$$\mathrm{e}_2 = \mathrm{LNS}\ F$$

$$\mathrm{P}_1 = [\mathrm{I}|0] \quad \mathrm{P}_2 = [[\mathrm{e}_2]_\times F + \mathrm{e}_2\mathrm{v}^{\mathrm{T}}|\lambda\mathrm{e}_2]$$

is the **most general camera pair** in canonical form, that is compatible with F